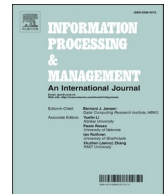




Contents lists available at ScienceDirect

Information Processing and Management

journal homepage: www.elsevier.com/locate/infoproman

An overview of opinion polarization: models, drivers, and strategic solutions

Shenghua Liu^a, Zhibin Wu^{a,*}, Luis Martínez^b^a Business School, Sichuan University, Chengdu 610065, China^b Department of Computer Science, University of Jaén, Jaén 23071, Spain

ARTICLE INFO

Keywords:

Polarization
Opinion dynamics
Social media
Echo chambers
Strategic solutions

ABSTRACT

In the context of social media and algorithmic personalization shaping information flows, opinion polarization poses a significant challenge to information processing and decision making. In this paper, we retrieved Web of Science records and applied a multi-stage screening process, yielding 145 rigorously selected papers on opinion polarization. From these sources, we developed a three-dimensional classification framework categorizing polarization models into individual behavior, group dynamics, and network structure. Our analysis reveals that cognitive bias in information processing, identity homophily within social groups, and algorithmic filtering in online platforms serve as core drivers of digital opinion polarization. Based on these insights, we propose strategic solutions across four interdependent domains: information level interventions, dialogue level facilitation, technology level adjustments, and psychology level guidance. We illustrate how these measures can be operationalized to mitigate echo chamber effects and foster cross-cutting engagement. This review synthesizes existing research to offer a structured foundation for understanding complex polarization mechanisms. It also provides a theoretical basis for future cross-platform dynamic modeling and policy development. Finally, we identify critical research gaps and outline future directions, highlighting the need for adaptive AI-driven moderation strategies, dynamic polarization models, and coordinated cross-platform policy interventions in the evolving digital landscape.

1. Introduction

With the rapid advancement of digital technologies, social media has fundamentally reshaped how individuals interact, share information, and form opinions on a global scale (De Marzo et al., 2020; Wu et al., 2022a; Cai et al., 2023; Guo & Hu, 2023; Shahbazi & Bunker, 2024; Ghenai et al., 2025). This shift has led to the phenomenon of opinion polarization, a complex social dynamic that affects political stability, social cohesion, identity formation, democratic engagement, and conflict resolution (Cinelli et al., 2021; Levin et al., 2021; Stewart et al., 2021; Kim & Rojas, 2025).

Opinion polarization can be understood from two complementary dimensions: as a dynamic process and as a static state. As a process, opinion polarization refers to the evolution in which opinions of individuals or groups become more extreme under the influence of factors such as social identity and algorithmic information exposure (Harel et al., 2020; Bliuc et al., 2021). This leads to a gradual increase in ideological and psychological distance. As a state, opinion polarization describes the distribution pattern of

* Corresponding author.

E-mail address: zhibinwu@scu.edu.cn (Z. Wu).

opinions within a society or group at a specific moment. It is often characterized by widening ideological divides, deteriorating intergroup trust, or a bimodal distribution of opinions (Lees & Cikara, 2021). Excessive polarization can undermine negotiation, compromise, and social cohesion, posing risks to social harmony and stability (Stewart et al., 2020; Baldassarri & Page, 2021; Dimant, 2024; Kadiyala & Shin, 2024). However, it is crucial to recognize that polarization is not inherently detrimental (Levendusky, 2010; Shi et al., 2019). In political contexts, for instance, a measured level of polarization can serve to clarify ideological distinctions, heighten public awareness of critical issues, and foster greater civic engagement.

Building on this cross-disciplinary clarification, it is also useful to specify the forms of polarization most relevant to this review, thereby ensuring conceptual precision and thematic consistency. Ideological polarization is the most direct manifestation, measuring how far apart different groups are on key policy issues within a political system (Dalton, 2006, 2021). A society with high ideological polarization means that people's opinions sharply opposed and lack consensus. This represents the most fundamental dimension for understanding opinion polarization. Affective polarization reveals its emotional dimension. Even when disagreements on a particular issue are minor, individuals may still develop intense negative feelings, such as aversion and distrust, for those on the opposing side (Druckman & Levendusky, 2019; Iyengar et al., 2019). This polarization is rooted in identity, which divides the world into us and them. This division emotionally intensifies the opposition that arises from differing opinions. Behavioral polarization, in turn, showcases the action-oriented dimension of opinion polarization, that is, how people act based on their opinions (Mason, 2013). Its core features are increasing partisan bias, political activism, and expressions of anger. This means that people may engage in hostile behaviors due to a strong sense of group belonging. The driving force is not necessarily the extremity of their opinions. Although this review adopts opinion polarization as the overarching theme, models, drivers, and interventions discussed herein often encompass mechanisms that span across these forms. Making these distinctions explicit provides a clearer conceptual anchor for the multidisciplinary analysis that follows.

Opinion polarization, as a complex social phenomenon, has garnered sustained attention over the years (Waller & Anderson, 2021; Kish Bar-On et al., 2024; Tao et al., 2025). The study of opinion polarization has evolved through three converging streams of research. First, social psychology established group polarization theory, revealing the internal mechanisms by which group discussion pushes individuals toward more extreme opinions (Stoner, 1961; Moscovici & Zavalloni, 1969). Subsequently, a new path emerged in the field of opinion dynamics, initially from models exploring how to reach consensus (French, 1956; DeGroot, 1974). By introducing key mechanisms such as bounded confidence and homophily, these models successfully simulated the process of social opinions from integration to division and opposition (Xia et al., 2011; Luo et al., 2014; Iyengar & Westwood, 2015), thereby providing a deducible theoretical framework for polarization research. Finally, the advent of the digital age sparked a third wave of research, deeply integrating the two preceding streams. The internet and social media not only amplified traditional polarization effects but also created information cocoons through algorithmic filtering, exacerbating opposition (Sunstein, 2001, 2002; Mutz, 2006). Subsequent research using computational social science methods has delved into the specific mechanisms of digital platform influence. For instance, studies have explored how algorithmic recommender systems can amplify polarization driven by biased assimilation (Dandekar et al., 2013). Others have investigated how diverse weak ties within social networks might, in turn, foster moderation (Barberá, 2014).

The past five years have further enriched our understanding of polarization through innovative methodologies and cross-disciplinary insights. Törnberg (2022) introduced "non-local interactions," demonstrating how digital media restructures social and ideological divisions along partisan lines, intensifying affective polarization by broadening exposure to partisan-aligned content beyond local contexts. Jost et al. (2022) presented a conceptual framework for understanding the cognitive-motivational mechanisms that drive political polarization, highlighting ego-justifying, group-justifying, and system-justifying motives as key factors. They argue that these psychological mechanisms, when interacting with social-communicative dynamics, reinforce ideological entrenchment and contribute to both symmetric and asymmetric polarization. Rathje et al. (2021) found that out-group animosity, rather than in-group favoritism, is the strongest predictor of social media engagement. Hostile content, they noted, significantly amplifies online interactions and reinforces divisive political discourse. The rapid evolution of digital technologies, particularly generative AI, has introduced new dimensions to polarization research (Gaitonde et al., 2021; Tokita et al., 2021; Vasist et al., 2024). Concurrently, researchers have begun developing strategic solutions to address polarization (Baliatti et al., 2021; Lechiakh et al., 2024).

1.1. Research type and scope

While existing research has extensively explored polarization's core theories, the impact of digital technologies, and the cognitive and social mechanisms at play, a comprehensive overview spanning these dimensions is still lacking. This paper therefore serves as a survey of opinion polarization research, drawing on insights from psychology, sociology, and technology studies to present a multidisciplinary perspective. By examining theoretical models, identifying key drivers, and evaluating strategic solutions, we offer a structured and cohesive overview of the field.

1.2. Research objectives

To fulfill the goals outlined in Section 1.1 and provide a structured, multidisciplinary survey of opinion polarization, this study sets out the following specific objectives:

- (1) Review and classify existing polarization models. Our aim is to identify, collect, and categorize significant models introduced in the past decade, grouping them into three broad dimensions: individual behavior, group dynamics, and network structure. By doing

so, we provide researchers with a unified taxonomy that highlights each model's theoretical underpinnings, assumptions, and limitations.

(2) Analyze the key drivers of opinion polarization from psychological, social, and technological perspectives. We intend to distill core factors such as cognitive biases, identity processes, emotional contagion, algorithmic recommendations, and feedback loops. For each driver, we examine how it operates in social media environments and interacts with other factors to exacerbate or mitigate polarization.

(3) Aggregate and evaluate multi-level strategic solutions for addressing polarization. This objective focuses on collecting intervention strategies reported in the literature, ranging from information level interventions to dialogue level facilitation, technology level adjustments, and psychology level guidance. We compare these solutions in terms of feasibility, effectiveness, and applicability across different platforms.

(4) Identify research gaps and propose future directions. Building on the preceding survey, we will pinpoint areas where current studies are limited, such as over-reliance on static models that ignore dynamic network evolution and cognitive biases; insufficient consideration of network and platform heterogeneity when analyzing drivers; narrow intervention strategies that lack multi-level integration and long-term empirical validation. We then outline concrete avenues for future work that can fill these gaps.

By achieving these objectives, our research not only offers a clear roadmap of what has been done in the field of opinion polarization but also provides actionable insights for researchers and practitioners to understand complex polarization dynamics and design effective solutions.

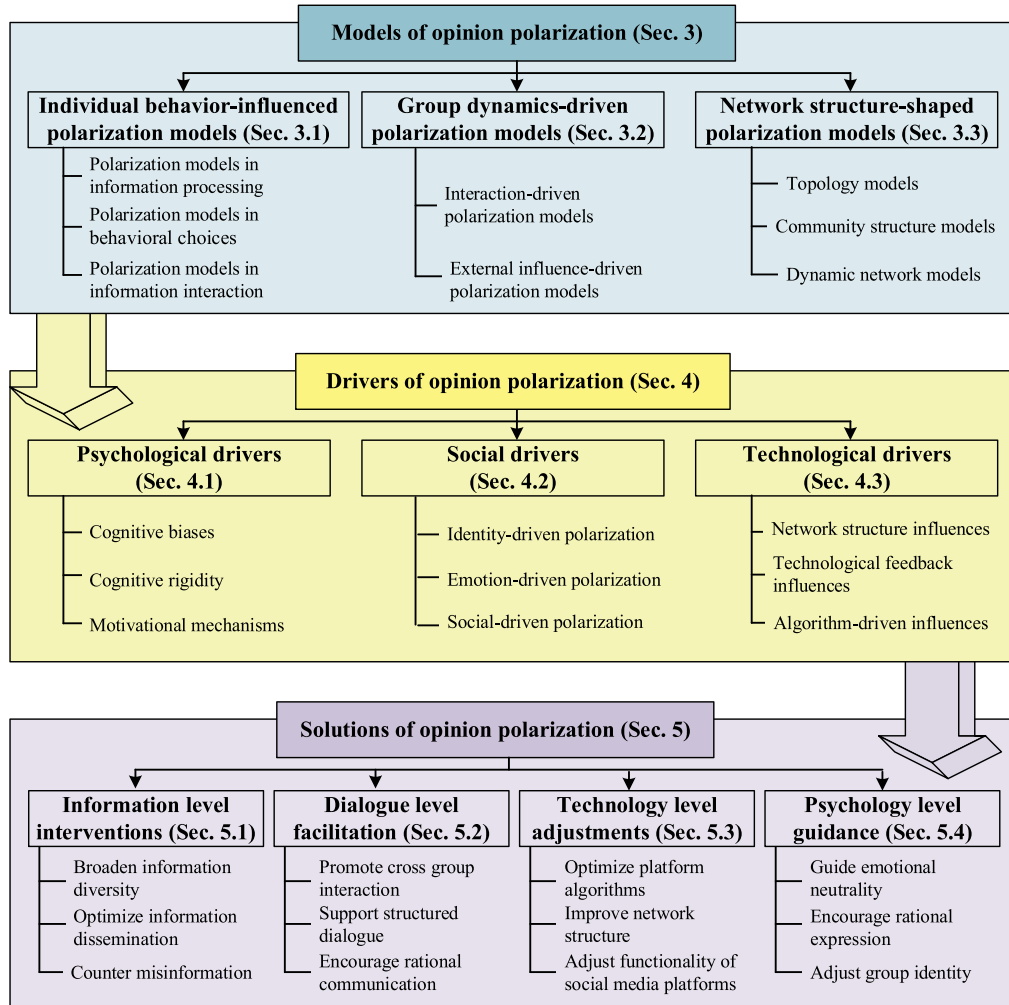


Fig. 1. Structure of the paper.

1.3. Organization

The structure of this paper is as follows: [Section 2](#) presents the research methodology. [Section 3](#) reviews models of opinion polarization. [Section 4](#) analyzes drivers behind polarization. [Section 5](#) discusses strategic solutions for addressing polarization. [Section 6](#) addresses open questions and emerging research directions. Finally, [Section 7](#) concludes the paper. To guide readers, [Fig. 1](#) illustrates the core chapter structure of the paper, providing a clear overview of the key topics discussed in each section.

2. Research methodology

This study conducts a systematic review to synthesize the existing body of knowledge on opinion polarization. The methodology was designed to be transparent, rigorous, and replicable, adhering to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines ([Caluwaerts et al., 2023](#); [Li et al., 2025](#)). This section outlines the multi-stage process, including the research questions that guided the review, the search and selection strategy, the criteria for inclusion and exclusion, and the procedure for data coding and analysis.

2.1. Research questions

This review analyzes the models, drivers, and strategic solutions of opinion polarization from a multidisciplinary perspective to construct a cohesive theoretical framework for the field. To guide this inquiry, we pose the following research questions (RQs):

- RQ1: What are the primary theoretical models and methods used to study opinion polarization?
- RQ2: How do these models characterize polarization, and what are their core conceptual and mathematical frameworks?
- RQ3: What are the main drivers of opinion polarization identified in existing research?
- RQ4: What are the representative strategies for mitigating or managing polarization?
- RQ5: What are the key limitations, gaps, and future challenges in opinion polarization research?

2.2. Literature search strategy

The initial literature search was conducted in the Web of Science (WoS) Core Collection, targeting the Science Citation Index Expanded (SCI-E) and Social Sciences Citation Index (SSCI) databases. WoS was selected for its rigorous indexing of high-impact, peer-reviewed journals, ensuring a high-quality foundation for our analysis across key disciplines like computer science and social sciences. The search was performed in late 2024, covering publications from 2014 to 2024 to focus on the contemporary era of digital media and algorithmic influence.

The following search query was used in the title, abstract, and keyword fields: TS=(“opinion polarization”) AND TS=(“polarization mechanism”) This query was intentionally designed for high precision to identify a core set of studies that directly investigate the mechanisms of polarization. To ensure comprehensiveness and high recall, we supplemented this database search with a manual citation snowballing process. Both forward and backward snowballing were conducted on the core articles to identify additional relevant studies, including foundational works published before 2014.

Table 1
Inclusion and exclusion criteria.

Criterion type	Criterion	Description
Inclusion criteria	Peer-reviewed publication	Included studies must be published in peer-reviewed journals or reputable, peer-reviewed conference proceedings.
	Thematic relevance	The study's primary topic must directly address the models, drivers, or solutions of opinion polarization, in alignment with the review's Research Questions (RQs).
	Methodological approach	The study must employ a clear analytical methodology, such as computational modeling, theoretical analysis, or empirical research (quantitative, qualitative, or mixed-methods).
	Novel contribution	The research contributes novel insights, methodological advancements, or theoretical perspectives that deepen the understanding of opinion polarization.
Exclusion criteria	Non-peer-reviewed materials	All non-peer-reviewed materials were excluded. This includes preprints (e.g., from arXiv), dissertations, theses, book chapters, editorials, book reviews, and other gray literature.
	Peripheral focus	Studies where opinion polarization was only mentioned as background context or a minor point, without being a central element of the analysis, were excluded.
	Redundancy or low novelty	To avoid duplication, redundant publications (e.g., retaining the final journal article over its preliminary conference version) were excluded. This criterion also applied to studies of low novelty that did not provide substantial new insights beyond the existing literature. By contrast, replication or validation studies conducted in different contexts were retained, as they contribute to robustness and generalizability.
	Lack of methodological rigor	Studies with significant methodological flaws, such as unclear data sources, non-transparent analytical procedures, or unsubstantiated conclusions, were excluded to maintain the overall quality of the review.

2.3. Inclusion and exclusion criteria

To ensure that the final selection of studies was characterized by high relevance, methodological rigor, and genuine contribution, we established a comprehensive set of inclusion and exclusion criteria. These criteria, detailed in Table 1, were developed to address key concerns in systematic reviews, such as ensuring publication quality, thematic relevance, and methodological soundness.

A fundamental requirement was the mandatory peer-review status of all included articles, leading to the systematic exclusion of non-peer-reviewed materials like preprints (e.g., from arXiv), dissertations, and other gray literature. Furthermore, each study was required to be thematically aligned with our research questions and employ a clear analytical approach, whether it be computational modeling, theoretical analysis, or empirical research. To resolve potential ambiguities in selection, we clarified our redundancy or low novelty criterion. It was designed to exclude duplicate publications. However, it did not apply to valuable replication or validation studies, which were retained for their contribution to the field's robustness. The addition of a criterion excluding studies with significant methodological flaws further reinforces the high-quality standard of this review. Table 1 provides a detailed summary of these rigorously applied criteria.

2.4. Screening process

The study selection was conducted through a multi-stage screening process in accordance with the PRISMA guidelines, as visually detailed in the flow diagram (Fig. 2). Each stage applied progressively stricter criteria to refine the dataset.

Identification: The process began by identifying records from multiple sources. An initial search of the WoS database yielded 244 articles, which were supplemented by an additional 60 records identified through citation snowballing. This resulted in a total of 304 records. After removing 17 duplicates, 287 unique articles were advanced to the screening stage.

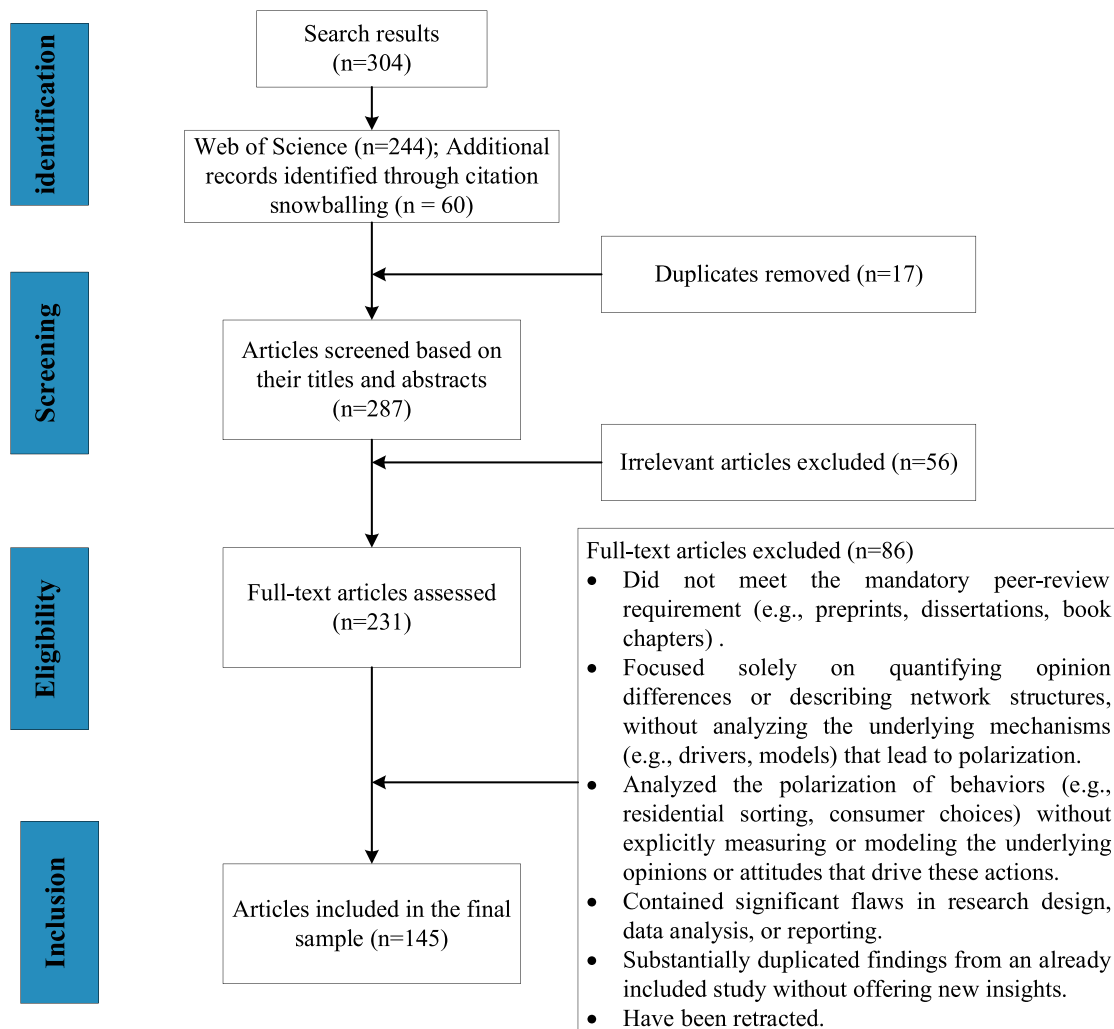


Fig. 2. PRISMA flow chart of the review.

Screening: The titles and abstracts of the 287 unique records were screened for relevance. During this stage, 56 articles were excluded as they were clearly outside the scope of the review, leaving 231 articles for full-text assessment.

Eligibility: The full texts of the remaining 231 articles were thoroughly assessed against the detailed inclusion and exclusion criteria outlined in Table 1. This rigorous evaluation led to the exclusion of 86 articles. Reasons for exclusion included: failure to meet peer-review standards, containing significant methodological flaws, being redundant, or having been retracted. Articles were also excluded if they focused solely on quantifying opinion differences without analyzing mechanisms, or if they analyzed behavioral polarization without linking it to underlying opinions.

Inclusion: Upon completion of the eligibility assessment, a final sample of 145 studies was included in the systematic review. This curated dataset forms the foundation for the analysis presented in the subsequent sections.

2.5. Data coding and analysis

To systematically extract and synthesize information from the final 145 articles, a structured coding procedure was developed and implemented. As no universally established coding framework existed for this specific multidisciplinary scope, we employed a thematic analysis approach.

Two authors independently coded a random subset of the articles (20%) to develop a coding framework. The framework was designed to extract key characteristics from each study, including:

- (a) General information: Publication year, discipline, and geographical context.
- (b) Polarization models: The type of model used (e.g., individual behavior, group dynamics, network structure), its core assumptions, and mathematical formulation.
- (c) Identified drivers: The psychological, social, or technological factors identified as drivers of polarization.
- (d) Strategic solutions: The type of intervention proposed or evaluated (e.g., informational, technological, psychological).

Any disagreements during the initial coding were resolved through discussion with a third author to reach consensus. The finalized coding framework was then applied to the remaining articles. This process ensured the data extracted from the literature directly addressed the research questions and allowed for a structured synthesis of the findings presented in the following sections.

2.6. Dataset description

Following the screening process described above, we included 145 high-quality studies on opinion polarization. To illustrate characteristics of the dataset, we conducted a quantitative analysis across three dimensions: temporal coverage, disciplinary distribution, and research type.

First, the temporal distribution of the sample is shown in Fig. 3. The included studies were primarily published between 2014 and 2024, exhibiting a significant upward trend in publication volume. This visually demonstrates the growing academic attention the topic of opinion polarization has received over the past decade, providing a solid empirical basis for this review. Second, the disciplinary distribution of the sample (as shown in Fig. 4) highlights the significant interdisciplinary nature of the field. Computer science forms the main body of research, closely followed by psychology, sociology, and political science. In addition, fields such as information science, physics, and communication studies also make important contributions, collectively constituting the multidimensional theoretical and methodological foundation of this research area and supporting the breadth of our interdisciplinary analysis. Finally, in terms of research type (as shown in Fig. 5), simulation and empirical studies are the two dominant methodologies, collectively accounting for nearly 90% of the total literature. Simulation research offers tools for theoretical deduction to explore polarization mechanisms, while empirical research provides real-world validation for the theories. Furthermore, theoretical reviews hold a

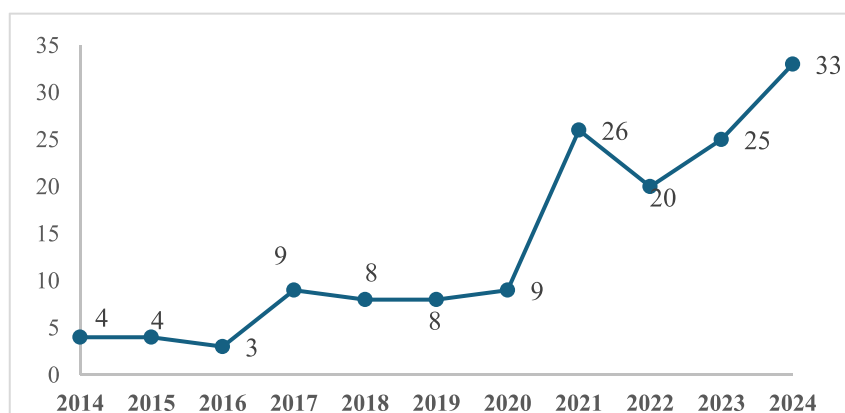


Fig. 3. Time distribution of the sample.

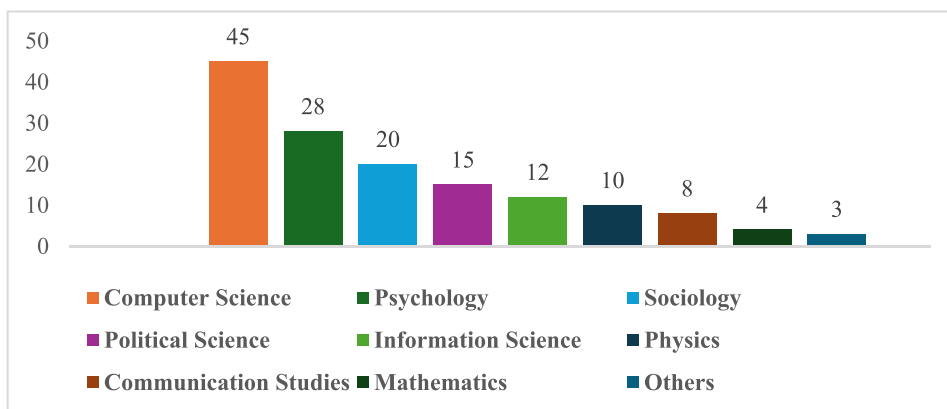


Fig. 4. Disciplinary distribution of the sample.

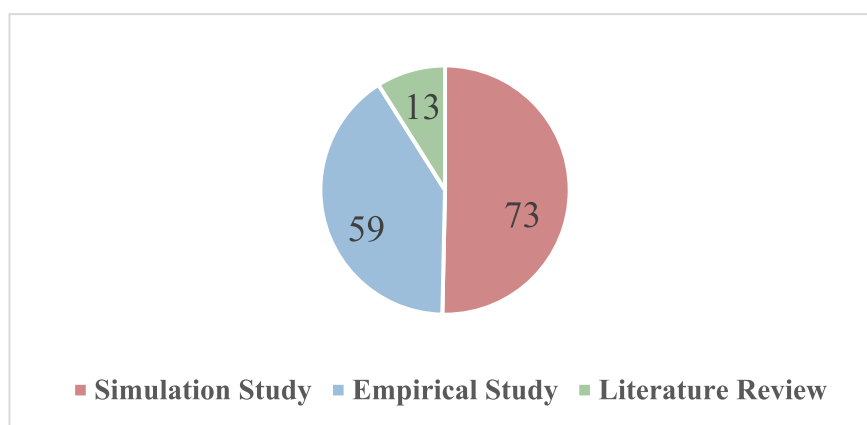


Fig. 5. Distribution of research types in the sample.

significant place, offering directional guidance and conceptual frameworks for the field's development.

This diverse dataset not only ensures the comprehensiveness of the analysis in the subsequent [Sections 3–5](#) but also provides a solid foundation for the discussion of future challenges and research directions in [Section 6](#).

3. Models of opinion polarization

This section reviews polarization models, focusing on the design principles and unique contributions from three perspectives: (i) the impact of individual behavior; (ii) group dynamics; and (iii) the shaping of network structure. This classification refers to the framework proposed by [Wen et al. \(2025\)](#), which emphasizes the roles of individual decision-making processes, social interactions, and network structures in shaping collective opinion dynamics. These perspectives form the basis for categorizing models, providing a multidimensional framework for understanding opinion polarization.

To lay a precise foundation for the models, it is important to first delineate the concept of polarization and its variants. In social psychology, group polarization originated from “risky shift” experiments ([Stoner, 1961](#)), later formalized by [Moscovici and Zavalloni \(1969\)](#). It mainly refers to the tendency for a group's opinions to divide and become more extreme through internal interactions. From opinion dynamics literature, however, polarization usually refers to one steady state in the spectrum from consensus to diversity ([Proskurnikov et al., 2015](#); [Mueller & Tan, 2018](#)), where the opinion extremalization is less emphasized. Another closely related concept is bi-polarization ([Mäs & Flache, 2013](#); [Proietti & Chiarella, 2023](#)), which refers specifically to the formation of two opposing opinion clusters that move toward opposite extremes.

[Fig. 6](#) outlines multidimensional framework for categorizing opinion polarization models, detailing three primary categories and their sub-dimensions. Individual behavior-influenced polarization models explore several aspects of individual choice. They examine how individuals selectively expose themselves to information, how their behavioral choices are shaped by social preferences and group dynamics, and how their interactions with external systems reinforce polarized opinions. Group dynamics-driven polarization models examine how polarization is amplified within groups, emphasizing the role of intragroup consistency in reinforcing shared beliefs and deepening intergroup divides. These models also highlight how social influence and reciprocal feedback between groups sustain and

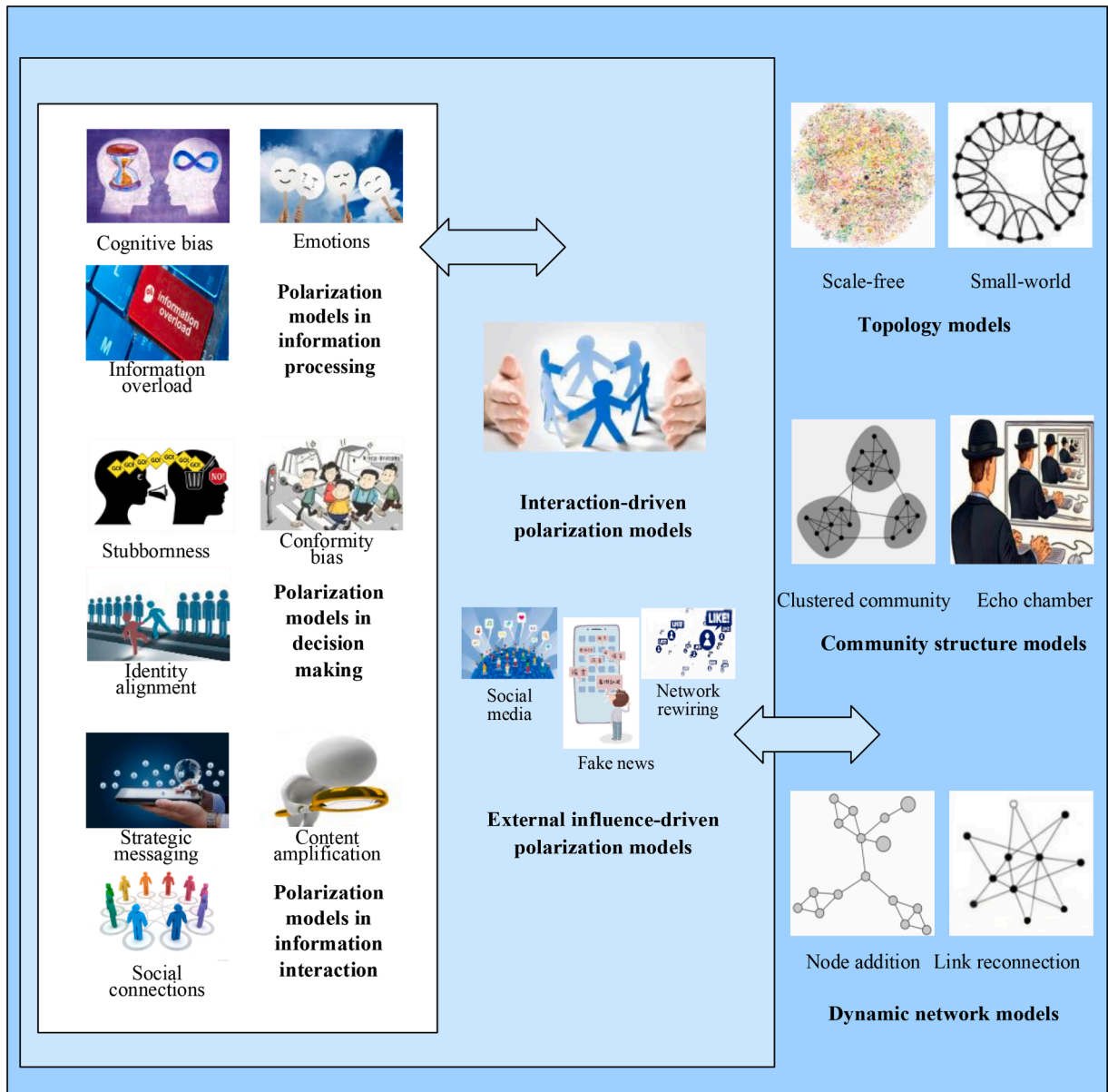


Fig. 6. A multidimensional framework of the polarization models.

escalate polarization, showing how social interactions propagate and exacerbate polarization across groups. Network structure-shaped polarization models investigate the topology of social networks, focusing on how structures like scale-free networks and small-world networks influence the formation of polarization. In particular, the clustering of like-minded individuals and the creation of echo chambers play a key role in intensifying polarization. These models also explore how dynamic changes in network relationships, such as node addition and link reconnection, affect the spread and strength of polarized opinions. This framework enables a comprehensive understanding of mechanisms that drive opinion polarization at the individual, group, and network levels.

For a detailed description of mathematical models, including their mathematical formulations and key references, please refer to [Table A1](#) in Appendix A. The differences in opinion polarization models arise from researchers' varying perspectives, assumptions, and modeling methods. Despite these differences, all models aim to simulate and clarify the mechanisms and dynamics driving polarization. To clarify conceptual distinctions between term pairs discussed, such as selective communication and selective interaction, readers are referred to [Table A2](#) in Appendix A.

3.1. Individual behavior-influenced polarization models

Here, we focus on how individuals process information, form opinions, and respond to personal beliefs and preferences. By examining key mechanisms such as information processing, behavioral choices, and information interaction, it uncovers the internal dynamics driving opinion polarization.

3.1.1. Polarization models in information processing

Here, models focus on investigating the internal cognitive mechanisms of information processing, which constitute the micro-foundations of opinion polarization at the individual level.

A primary research cluster focuses on how cognitive biases distort information processing. Computational models commonly treat confirmation bias, motivated reasoning, and the backfire effect as key drivers of polarization. These biases prompt individuals to maintain belief consistency through selective information processing, thereby exacerbating polarization (Sobkowicz, 2018; Zafeiris, 2022; Maksymov & Pogrebna, 2024). For instance, the representative study by Del Vicario et al. (2017) integrates confirmation bias into the bounded confidence model. Their work demonstrates how a preference for consistent information naturally leads a group toward division.

Secondly, continuous opinion and dynamic models focus on the information environment itself, exploring how features such as the quantity, frequency, and heterogeneity of interactions interplay with cognitive biases (Baron, 2021; Krishnarao & Selvamani, 2024). Hybrid models show that, far from fostering consensus, information overload or excessive exchange can exacerbate polarization by increasing cognitive load and reinforcing individuals' reliance on biases (Yang et al., 2022).

Furthermore, an emerging cluster shifts the focus from pure cognition to emotional factors. Recent game-theoretic and agent-based models emphasize the driving role of emotion. They demonstrate that an emotional attachment to beliefs can strengthen in-group opinion clustering. Meanwhile, the algorithmic amplification of hostile rhetoric can directly catalyze affective polarization (Quante et al., 2024; Sarkar & Hadfield, 2024).

3.1.2. Polarization models in decision making

Beyond cognitive processing, models in this section focus on how active behavioral choices, such as stubbornness, conformity, and identity alignment, serve as direct drivers of opinion polarization.

A primary research cluster focuses on intrinsic behavioral tendencies, particularly stubbornness and conformity. Voter models and agent-based simulations commonly assume that a subset of individuals holds fixed opinions, allowing for the study of how these few key nodes impact the entire network's opinion distribution (Das et al., 2014). For instance, the representative study by Arendt and Blaha (2015) used a zealotry-based model to demonstrate a key finding. They showed that even a few strategically positioned stubborn individuals are sufficient to sway an entire network and sustain long-term disagreement.

Further refining the understanding of conformity and social influence, another major cluster addresses behaviors conditioned by the social environment, such as feedback, learning, persuasion, and social preferences. Social reinforcement learning models show that positive peer feedback or multi-agent learning can reinforce an individual's existing opinions, thereby driving bi-polarization in modular networks (Banisch & Olbrich, 2019; Meylahn & Meylahn, 2024). Concurrently, other models explore different mechanisms. Persuasion models investigate how individual receptiveness to persuasion shapes interactive behavior, while social preference theory models examine how different trade-offs in gains affect collective polarization (Chen et al., 2020; Cao & Reed, 2025).

Furthermore, a crucial dimension is identity-driven opinion alignment. These models posit that individuals' behavioral choices are often made to align their opinions with those of their identity group. Augmenting the classic Hegselmann-Krause model with identity constructs, Kalvas et al. (2022) revealed how polarization emerges as an endogenous phenomenon. This happens when individuals adjust their opinions to reinforce their identity, which in turn forms entrenched opinion clusters resistant to external influence. Concurrently, new explorations have emerged in modeling methodology. These include the introduction of geometric dynamics models and non-linear processing methods, which provide new perspectives and mathematical tools for analyzing and addressing polarization (Gaitonde et al., 2021; García-Zamora et al., 2021).

3.1.3. Polarization models in information interaction

Here, we explore the interaction process between individuals and external information systems, exploring how factors such as strategic information, personalized content, and social connections shape opinions.

Multidimensional opinion dynamics models can capture the interdependence of opinions. For example, extensions to the classic Friedkin-Johnsen model provide a more realistic framework for understanding polarization on complex issues by depicting how individuals' opinions on multiple related topics co-evolve (Parsegov et al., 2017). A core focus of current research is the shaping role of personalized information environments. Augmented dynamic and differential equation models reveal that these systems can significantly accelerate polarization. This occurs when personalized content selection combines with users' cognitive biases, leading to the rapid formation of homogeneous communities (Bhalla et al., 2023; Valensise et al., 2023; AlMahmoud & Mallmann-Trenn, 2024).

Another cluster focuses on social amplification effects during the interaction process. Related content amplification, enhanced cascade, and geometric models respectively reveal how the individual pursuit of attention, group polarization effects, and external strategic information jointly amplify opinion extremity (Dai et al., 2022; Lim & Bentley, 2022; Hazla et al., 2024). Finally, cutting-edge models examine how individuals dynamically adjust their network connections based on evolving opinions. These models illustrate a dynamic feedback loop. On one hand, individuals are influenced by the network. On the other hand, their decisions on connection and sharing in turn reshape the network. This feedback loop can lead to persistent, long-term polarization (Bolletta & Pin, 2025).

3.2. Group dynamics-driven polarization models

This section explores how group interactions and social influences drive opinion polarization by reinforcing internal cohesion and deepening inter-group divides. It explores polarization through internal dynamics within groups and the impact of external influences on group behavior.

3.2.1. Interaction-driven polarization models

This section focuses on interaction processes within groups, exploring how selective communication and social influence between individuals can engender and reinforce polarization.

Classic social influence models laid the foundation for this field, with a core mechanism being the tendency of individuals to be influenced by their neighbors. Represented by the model of Axelrod (1997), this line of research reveals how local convergence can lead to global cultural divergence and polarization at the macro level. These early group interaction dynamics models emphasize that discussion within a group is itself sufficient to push collective decisions toward extremes (Rao & Steckel, 1991; Flache et al., 2017; Galam, 2023).

Another mainstream cluster is built on the assumption of bounded confidence, where individuals interact only with like-minded others. By setting trust thresholds or similarity boundaries, these dynamic models naturally explain the formation, separation, and even opposition of opinion clusters, and can be extended to large-scale, multi-group scenarios (Li et al., 2013; Bakaryan et al., 2025; Cyranka & Mucha, 2024). In contrast to models focusing on influence, argument-based communication models delve deeper into the content of interactions. These models posit that polarization stems from individuals repeatedly exchanging and being exposed to one-sided yet novel persuasive arguments within a group, which reinforces their initial tendencies. Core variables in these models include argument strength and informational biases (Mäs & Flache, 2013; Proietti & Chiarella, 2023).

Finally, cutting-edge research integrates more complex interaction factors through multidimensional and networked models. These models can simultaneously consider multiple mechanisms such as social influence, homophily, radicalization, topic interconnectedness, and even noise or miscommunication. They reveal how the superposition and coupling of these various factors in more realistic scenarios jointly drive and sustain a group's polarized state (Turner & Smaldino, 2018; Baumann et al., 2020, 2021; Gubanov et al., 2021; Biondi et al., 2023).

3.2.2. External influence-driven polarization models

Building on the internal interactions discussed previously, models in this section further explore how external influences, such as specific social media mechanisms, algorithmic filtering, and strategic information, shape and intensify group polarization from the outside.

Specific mechanisms of social media platforms are a primary source of external influence. Agent-based models (ABMs) and extensions to the Friedkin-Johnsen model reveal that specific on-platform behaviors, such as network rewiring, retweets, and friend-of-friend recommendations, can amplify in-group homogeneity and accelerate opinion radicalization (Rychwalska & Roszczynska-Kurasinska, 2018; Bhalla et al., 2023; Das et al., 2023). Algorithmic content curation is another key external driver. Related ABMs show that algorithm-generated filter bubbles and personalized content recommendations systematically exacerbate collective polarization. They do this by limiting a group's exposure to heterogeneous information, which in turn shapes isolated echo chambers at the macro level (Keijzer et al., 2024).

Finally, a core cluster focuses on the external injection of strategic or false information. Various dynamic and mathematical models explore how external forces, such as dynamic interactions, the dissemination of fake news, time-varying external signals, elite manipulation of media asymmetries, and the alignment of issues along ideological spectrums, can serve as a powerful means of intervention to actively create and deepen societal rifts and group antagonism (Franceschi et al., 2023; Mobilia, 2023; Macias & Vera, 2024; Schweighofer & Garcia, 2024).

3.3. Network structure-shaped polarization models

This subsection explores how network structures influence opinion polarization, focusing on the impact of global topology, local community dynamics, and network evolution. By progressing from static structural properties to dynamic changes, these types of models provide a comprehensive understanding of polarization in networked environments.

3.3.1. Topology models

Here, we review how network structures influence opinion dynamics and polarization. These models explore how the static structural features of a network, such as connection patterns and relational signs, provide the conditions for polarization to emerge and be maintained.

Opinion dynamics models on specific topologies reveal that a network's macro-level structure decisively influences opinion propagation. Research shows that different topological features can lead to an uneven distribution of information, which in turn can engender or consolidate opinion polarization. Examples of such features include the hubs in scale-free networks, the asymmetry of information flow in directed networks, and the community structures in stochastic block models (O'Connor & Weatherall, 2018; Ye et al., 2022; AlMahmoud & Mallmann-Trenn, 2024; Brooks et al., 2024; Fraiman et al., 2025).

A particularly important research cluster involves signed networks, which provide a more realistic depiction of social structures by introducing positive/negative relationships. Research based on affine dynamics and extended DeGroot models finds that a network's

internal drive toward structural balance (i.e., the friend of a friend is a friend) naturally splits the network into two mutually hostile, polarized camps (Cisneros-Velarde et al., 2021; Li & Tang, 2023; Shi et al., 2023). This provides a powerful topological explanation for social antagonism.

Finally, other models investigate how specific interaction rules on a given static topology lead to polarization. For example, bounded confidence and kinetic models show that opinion fragmentation can occur even when the network structure is fixed. This happens if individuals have limited information horizons or follow specific statistical connectivity distributions, leading to persistent polarized states (Gubanov & Petrov, 2019; Loy et al., 2022; Pérez-Martínez et al., 2023).

3.3.2. Community structure models

Models in this category focus on the network's meso-level, particularly community structure and selective interactions. They concentrate on the formation mechanisms of echo chambers and homogeneous communities. Furthermore, they examine how these structures amplify polarization.

A core cluster is dedicated to revealing the formation and reinforcement mechanisms of echo chambers. Various influence and cascade models indicate that selective exposure at the individual level and algorithmic bias at the platform level jointly promote homogeneous clustering (Sirbu et al., 2019; Prasetya & Murata, 2020; Valensise et al., 2023). For example, the representative study by Sasahara et al. (2021) clearly demonstrates through a social influence model that the active behavior of selective unfollowing significantly accelerates echo chamber formation and deepens opinion segregation.

Another line of research focuses on differentiated interaction rules inside and outside of communities. Research based on voter and Schelling-based models finds that the combination of high intra-community conformity and inter-community repulsion or distrust is a key driver for the long-term persistence of polarized opinions. This pattern of distinct internal versus external interaction, coupled with competition among media or network fragmentation, further solidifies community boundaries (Quattrociocchi et al., 2014; Feliciani et al., 2017; Weron & Szwabinski, 2022).

Cutting-edge models are beginning to introduce strategic or game-theoretic perspectives to study polarization within community structures. Game-theoretic, co-evolutionary, and extended dynamic models explore how, within established community structures, connection preferences can either bridge or exacerbate community division, and how external actors can manipulate the entire community's opinion trajectory by strategically influencing key individuals (Matakos et al., 2017; Dai et al., 2022; Borges et al., 2024; Sarkar & Hadfield, 2024; Zhang et al., 2024).

3.3.3. Dynamic network models

Finally, this section explores the highest level of complexity in network analysis: the co-evolution of opinions and the network structure itself. Here, the network is no longer a static background but a dynamic entity that interacts and co-evolves with opinion dynamics (Parsegov et al., 2017).

The core mechanism of mainstream co-evolutionary models is that while individuals' opinions are influenced by the network structure, their opinions and preferences in turn drive adjustments to that structure. These models simulate processes such as link rewiring, relationship strength adjustment, and confirmation bias-based friend recommendations. They reveal that the positive feedback loop between opinions and the network is key to forming and solidifying polarization. For example, the representative study by Bolletta and Pin (2025) establishes a framework where individuals simultaneously adjust their opinions and network connections. This work clearly demonstrates how such bidirectional feedback can lock a society into a highly polarized structure (Yuan et al., 2020; Bhalla et al., 2023; Sobkowicz, 2023).

Beyond opinion-driven adaptive adjustments, other models introduce system-level dynamic factors. Epidemic-based and algorithmic dynamic models, for instance, highlight important external variables that can reshape network connections and influence polarization. These variables include the entry and exit of network members and the platform's content distribution algorithm. These studies indicate that polarization is not only a product of micro-level homophilic choices but also deeply influenced by macro-level platforms and population dynamics (de Arruda et al., 2022; Yuan et al., 2022).

The classification of polarization models into individual behavior-influenced, group dynamics-driven, and network structure-shaped dimensions provides a foundational understanding of this complex phenomenon. While these categories help to analytically separate the core mechanisms, the most advanced research demonstrates that polarization emerges from the dynamic interplay between these levels, often mediated by technology. Rather than operating in isolation, cognitive biases at the individual level are amplified by group interactions, which are in turn constrained and shaped by the network's structure and the algorithms governing it. For instance, recent studies provide a more holistic view by explicitly integrating these dimensions. Some models have combined individual factors like confirmation bias with group influence dynamics and algorithm-driven link recommendations (Bhalla et al., 2023), revealing how these mechanisms collectively accelerate polarization. Other integrative approaches have shown how platform algorithms co-evolve with dynamic network structures to foster echo chambers (de Arruda et al., 2022), or how algorithmic bias embedded within social interaction models can systematically fragment the opinion landscape (Sirbu et al., 2019). This clear shift toward integrative models that bridge individual, group, and network perspectives offers a more robust and realistic framework. Through analyzing these increasingly sophisticated models, we can better comprehend the multifaceted drivers of opinion polarization and gain stronger theoretical insights for the formulation of strategic solutions.

4. Drivers of opinion polarization

The formation of opinion polarization is a complex, multifaceted process shaped by the interplay of psychological, social, and

technological factors. To gain a deeper understanding of this phenomenon, this section examines how each of these dimensions contributes to the intensification of polarization from the perspectives of psychology, sociology, and technology. Key drivers are presented in Table 2. The interaction of these psychological, social, and technological drivers collectively propels individuals and groups towards more extreme positions across social, political, and cultural issues. In the following subsections, we will explore these drivers in further detail, discussing the theoretical frameworks that underpin their role in the polarization process.

4.1. Psychological drivers

The psychological drivers of opinion polarization primarily influence the formation and reinforcement of individuals' positions and attitudes through their thinking patterns, emotional needs, and psychological mechanisms. To better understand how these factors exacerbate polarization, they can be categorized into three main groups: cognitive biases, cognitive inflexibility, and motivational mechanisms. These categories are based on individuals' varying responses in information processing, emotional reactions, and internal needs, explaining how psychological mechanisms contribute to the development of polarization.

Cognitive biases refer to systematic errors in information processing that distort how individuals interpret and evaluate information, leading to a reinforcement of pre-existing beliefs and greater polarization (Azzopardi, 2021). For instance, through theoretical modeling, Dandekar et al. (2013) simulated how biased assimilation and homophily could jointly promote interactions among similar opinions and intensify societal polarization. Another simulation-based study by Mueller and Tan (2018) highlighted the role of belief consistency, proposing a model where the need for conceptual coherence drives opinion divergence and group polarization. Moving beyond pure simulation, Banisch and Shamon (2025) integrated empirical research with computational modeling, showing how cognitive biases, like favoring consistent arguments, drive both individual and collective polarization. By reviewing and synthesizing existing empirical literature, Fernbach and Van Boven (2022) argued that false polarization, fueled by cognitive biases like categorical thinking and emotional amplification, amplifies perceived intergroup differences. These biased cognitive processes distort reality,

Table 2
Drivers of opinion polarization.

Category	Drivers and Description	Refs.	Evidence Type
Psychological drivers	Confirmation bias: A cognitive tendency to favor information that reinforces existing beliefs. Reduce openness to opposing opinions.	(Del Vicario et al., 2017)	Simulation study
		(Sobkowicz, 2018)	Simulation study
		(Bhalla et al., 2023)	Simulation and empirical studies
Social drivers	Backfire effect: Exposure to contradicting information strengthens original beliefs. Deepen resistance to opposing opinions.	(Maksymov & Pogrebna, 2024)	Simulation study
	Motivated reasoning: Individuals interpret information based on emotional or cognitive needs, favoring supportive interpretations.	(Rogowski & Sutherland, 2016)	Empirical study
	Selective exposure: Individuals choose information that aligns with their existing beliefs, avoiding contradictory opinions.	(Sobkowicz, 2018)	Simulation study
	Social identity theory: Individuals define themselves based on group memberships. Ingroup favoritism and outgroup hostility emerge, deepening divisions between groups.	(Prasetya & Murata, 2020)	Simulation study
		(Rathje et al., 2021)	Empirical study
	Social influence theory: Individuals' beliefs and behaviors are shaped by social norms. Group pressures push individuals toward more extreme positions, intensifying polarization.	(Jost et al., 2022)	Literature review
		(Kalvas et al., 2022)	Simulation study
		(Del Vicario et al., 2017)	Simulation study
	Social preference theory: Individuals are influenced by the preferences and behaviors of others in their social group.	(Flache et al., 2017)	Literature review
		(Mobilia, 2023)	Simulation study
Technological drivers	Social comparison theory: Individuals evaluate their opinions by comparing them to others.	(Chen et al., 2020)	Simulation and empirical studies
	Social judgment theory: Individuals' acceptance of new information is influenced by their prior beliefs.	(Wang et al., 2024)	Empirical study
	Opinion amplification: Social interactions, especially on platforms like social media, amplify individuals' opinions.	(Sieber & Ziegler, 2019)	Empirical study
	Recommendation algorithm: Algorithms suggest content based on past behavior, reinforcing existing preferences.	(Liu et al., 2023a)	Simulation study
		(Lim & Bentley, 2022)	Simulation study
		(Sirbu et al., 2019)	Simulation study
		(de Arruda et al., 2022)	Simulation study
		(Bhalla et al., 2023)	Simulation and empirical studies
		(Valensise et al., 2023)	Simulation and empirical studies
	Structural balance: Balanced relationships contribute to stability. Imbalances create tension, pushing individuals toward more extreme positions.	(Li & Tang, 2023)	Simulation study
		(Shi et al., 2023)	Simulation study

deepening polarization even when ideological differences are minimal.

Cognitive rigidity refers to mental inflexibility that strengthens preexisting beliefs, hinders adaptation to new information, and deepens polarization (Zmigrod, 2020). In their theoretical framework, Wu et al. (2022b) highlight the relationship between cognitive rigidity and polarization, suggesting that individuals and groups, lacking cognitive flexibility, reinforce existing beliefs when facing complex issues. This reinforcement leads to opinion polarization and exacerbates social division. Through a comprehensive literature review, Jost et al. (2022) further expanded on this idea by identifying how cognitive rigidity is fueled by ego-justification, group-justification, and system-justification, which promote the entrenchment of polarized opinions. Providing empirical evidence, Xie et al. (2023) demonstrated through a machine learning analysis of real-world social media data that individual characteristics like curiosity and emotional stability significantly influence polarization around vaccine hesitancy. Their findings highlight how cognitive rigidity can manifest, causing individuals to cling to preexisting beliefs and thus intensifying polarization.

Motivational mechanisms are psychological drivers that shape attitudes and exacerbate polarization (Jost et al., 2022). Through a quantitative analysis of national survey data, Drummond and Fischhoff (2017) demonstrated that higher science literacy can amplify polarization on controversial topics, likely due to identity-protective reasoning. Using a controlled experimental design, Guilbeault et al. (2023) further showed that while unrecognized cognitive differences can promote belief change, once these differences are revealed, motivational mechanisms like group cohesion can trigger resistance and intensify polarization. Providing a broad theoretical overview, Jost et al. (2022) synthesized existing research in a comprehensive review, arguing that motivations such as group-justification and system-justification are central drivers of polarization. This is because they reinforce loyalty to one's ideological ingroup.

By reinforcing existing beliefs and filtering information, cognitive biases spark polarization and lay the groundwork for cognitive rigidity and motivational mechanisms. Cognitive rigidity results from reinforced biases, while motivational mechanisms amplify polarization. These three factors create a psychological loop: biases and rigidity strengthen beliefs, and motivation drives loyalty to those beliefs, resisting opposing opinions and intensifying polarization.

4.2. Social drivers

The social drivers of opinion polarization primarily stem from individuals' behavior within groups, emotional expression, and social interactions. To better understand how these factors exacerbate polarization at the societal level, social drivers are divided into three categories: identity-driven polarization, emotion-driven polarization, and social-driven polarization. These categories are based on the different manifestations of group identity, emotional responses, and social network structures, explaining how social mechanisms contribute to the deepening of polarization.

Identity-driven polarization emphasizes how group identity and identification amplify ideological divides (Bliuc et al., 2024). As individuals align their opinions with their group, they view opposing groups as adversaries, deepening polarization. This is particularly evident in political and social contexts, where stronger in-group bonds and exclusion of opposing opinions intensify opinion divergence and social fragmentation. Empirical research provides strong evidence for this. For example, by analyzing large-scale survey data and conducting a survey experiment, Rogowski and Sutherland (2016) showed that ideological divergence between political candidates heightens affective polarization. Likewise, in their quantitative analysis of national election data, Webster and Abramowitz (2017) argued that affective polarization is fueled by growing ideological divisions. Theoretical modeling offers complementary insights. Kalvas et al. (2022) studied how group identity dynamics amplify polarization, driving opinion alignment with group identity.

Emotion-driven polarization highlights how emotional and ideological differences fuel polarization (Webster & Albertson, 2022). Emotional expressions like anger, fear, and pride strengthen commitment to ideological positions and foster negative feelings toward opposing groups. This process deepens ideological divides and intensifies social divisions. In a multi-method study combining large-scale data analysis and controlled experiments, Rathje et al. (2021) demonstrated that out-group animosity, driven by intergroup conflict, significantly increases engagement on social media platforms like Facebook and Twitter. This dynamic amplifies political polarization by promoting the spread of divisive content. Similarly, through a series of online experiments, Sun et al. (2023) highlighted the role of content ideology, where emotionally charged sentiments in social media posts contribute to opinion polarization. They found these dynamics are further moderated by symbolic expressions, such as likes and shares, which amplify the emotional impact and reinforce ideological divides.

Social-driven polarization is closely tied to social networks and interaction structures. Social media platforms and network dynamics, particularly homogeneity and group formation, amplify polarization. These networks expose individuals to like-minded opinions, reinforcing echo chambers and limiting access to diverse perspectives. Algorithmic filtering and recommendations further strengthen group cohesion, deepening divisions between opposing groups. Evidence from experimental research illustrates these micro-level processes. For instance, through three controlled experiments, Sieber and Ziegler (2019) proposed a processing effort account where social comparison and persuasive argumentation drive polarization. Similarly, in a series of online experiments, O'Reilly et al. (2024) demonstrated that even novel opinion sharing can trigger opinion-based identification and reinforce group dynamics. Moving from controlled settings to real-world behavior, Wang et al. (2024) conducted a quasi-experiment on an online review platform. Their analysis found that being friended by another user increased review polarization, suggesting that normative social influence in this context encourages more extreme, identity-signaling behavior.

Identity-driven polarization underpins both emotion-driven and social-driven polarization. Emotions reflect identity alignment, intensifying group hostility. Social-driven polarization amplifies these dynamics through information filtering and social media effects. These factors magnify identity-based and emotional divides, deepening societal polarization.

4.3. Technological drivers

The technological drivers of opinion polarization primarily arise from the interactions between algorithms, network structures, and information flow within digital media platforms and social networks. These technological factors contribute to the intensification of ideological divides and the reinforcement of social divisions. They can be categorized into three main types: network structure influences, technological feedback influences, and algorithm-driven influences. Each element exacerbates opinion divergence through distinct mechanisms, intensifying polarization in society.

Network structure influences shape group dynamics by fostering homophilic clustering, where individuals associate with others who share similar beliefs. This process reinforces existing opinions and deepens ideological polarization (Kwon et al., 2017). Empirical analyses of social media data consistently reveal this pattern. For instance, by mapping the network of climate change discussions on Twitter, Williams et al. (2015) found that homophilic echo chambers amplify polarization. In another empirical study of Twitter, Du and Gregory (2016) quantified this effect, showing how network dynamics limit exposure to diverse opinions. Expanding this analysis across multiple platforms, Cinelli et al. (2021) confirmed that homophilic clustering is a key driver of polarization online. Theoretical modeling provides a complementary perspective: Liu et al. (2023b) proposed a co-evolving network simulation showing that the interaction between opinion homogenization and homophily clustering naturally leads to stable, bipolarized structures.

Technological feedback influences describe how digital platforms and network structures create self-reinforcing loops, amplifying polarization. Users engage with content that aligns with their beliefs, and algorithms amplify this content, deepening ideological divides and reinforcing social divisions. Using an ABM, Bessi et al. (2016) demonstrated that online networks foster polarization through homophily, creating a feedback loop where users interact with content that reinforces their beliefs. In a large-scale field experiment on Twitter, Bail et al. (2018) showed that exposure to opposing opinions can intensify polarization through backfire effects, illustrating a technological feedback loop where algorithmic exposure to contrary opinions deepens divides instead of bridging them. Finally, in a critical review of the existing literature, Kreiss and McGregor (2024) argued that digital platforms encourage partisan identity performance, creating a feedback loop where engagement with polarized content deepens social divisions.

Algorithm-driven influences refer to how personalized content delivery algorithms shape user experiences on digital platforms. By curating content based on past behavior and preferences, these algorithms narrow exposure to opposing opinions and reinforce engagement with ideologically aligned content. Using an agent-based bounded confidence model, Sirbu et al. (2019) found that algorithmic bias intensifies polarization by promoting interactions among like-minded individuals. In a hybrid study combining theoretical modeling with an empirical analysis of Twitter data, Berman and Katona (2020) showed that curation algorithms amplify polarization by limiting exposure to diverse opinions and boosting engagement with like-minded content, creating filter bubbles. Based on a two-wave panel survey, Feezell et al. (2021) found that algorithm-driven news delivery amplifies engagement through tailored content, but indirectly reinforces polarization by limiting exposure to diverse perspectives. From another modeling perspective, Santos et al. (2021) showed in their agent-based simulation that link recommendation algorithms foster polarization by creating ideologically homogenous communities. In a large-scale field experiment, Levy (2021) demonstrated that social media algorithms amplify political polarization by limiting exposure to opposing opinions. Finally, using a computational model tested against longitudinal survey data, Törnberg (2022) argued that digital media amplify affective polarization by aligning diverse conflicts along partisan lines.

Network structure influences provide the foundation for group division, while technological feedback influences accelerate the intensification of polarization within these structures. Algorithm-driven influences further deepen information homogeneity and intra-group polarization through personalized recommendations. Together, these factors exacerbate ideological divides in society and hinder cross-group communication and understanding, profoundly influencing opinion polarization.

This section has reviewed the psychological, social, and technological drivers that contribute to opinion polarization. However, a deeper understanding requires looking beyond these categories to see how the drivers interact. The true force of polarization lies not in any single factor, but in their combined and reinforcing effects. They often form powerful feedback loops where each type of driver strengthens the others, creating a self-reinforcing cycle of division. A prime example of this interplay is selective exposure. While its roots are psychological, rooted in an individual's motivation to seek out confirming information, its consequences are profoundly social and technological. For example, technological drivers, like recommendation algorithms, create personalized information environments. These environments encourage social drivers, such as the tendency for like-minded people to cluster together, leading to the formation of echo chambers (Cinelli et al., 2021; Santos et al., 2021). Within these technologically-structured social spaces, individuals' psychological drivers, particularly cognitive biases like confirmation bias and biased assimilation, are powerfully activated and reinforced due to the lack of diverse perspectives (Dandekar et al., 2013; Del Vicario et al., 2017). This cycle is strengthened when emotional content triggers hostility toward opposing groups. Because such content generates high engagement, algorithms tend to amplify it, further deepening the psychological and social divides (Rathje et al., 2021). Understanding this interplay is therefore essential. It shows that effective solutions, which will be discussed next, must tackle this complex system as a whole, rather than focusing on any single factor in isolation.

5. Strategic solutions of opinion polarization

Opinion polarization is a complex and multi-dimensional phenomenon, influenced not only by how information is disseminated but also by social interactions, technological design, and individual psychological processes (Arora et al., 2022). To effectively address this issue, this section analyzes strategic solutions for opinion polarization, categorizing them into four key dimensions: information level interventions, dialogue level facilitation, technology level adjustments, and psychology level guidance. Each dimension tackles

different aspects of polarization, including enhancing information diversity, fostering rational dialogue, and optimizing algorithms. The overarching goal is to promote emotional neutrality and create a more inclusive, balanced, and empathetic environment.

5.1. Information level interventions

Information level interventions aim to enhance the quality and diversity of information within social networks to reduce bias and echo chamber effects, which are key drivers of polarization. These strategies focus on broadening information exposure, balancing perspectives, and countering misinformation, ultimately fostering a more inclusive and less polarized environment.

One effective approach is to humanize political figures and expose individuals to diverse perspectives. For example, Rogowski and Sutherland (2016) through a rigorous empirical study, found that sharing biographical information about public officials can effectively mitigate the polarizing effects stemming from ideological differences. Similarly, Matakos et al. (2017) used a simulation study to explore how algorithms can identify key nodes, aiming to break down information bubbles by promoting education and exposure to diverse opinions. Likewise, the empirical study by Van der Linden et al. (2017) focused on misinformation regarding climate change. Their survey experiment demonstrated that communicating the scientific consensus and using a preemptive refutation strategy can help the public understand scientific facts. This approach reduces the impact of misinformation and curbs its spread in polarized discussions. Building on this foundation, Judge et al. (2023) in a comprehensive literature review, expanded on these ideas. They systematically summarized and recommended reframing environmental issues to align with the audience's values. They argued that using trusted messengers can increase credibility, reduce ideological resistance, and enhance the effectiveness of information in mitigating polarization.

In addition to these educational approaches, recent studies have also explored the role of tools and algorithmic changes in fostering more balanced information flows. A simulation study by Lim and Bentley (2022) emphasized the need to inject non-polarized, evidence-based opinions into networks to counteract amplification of extreme opinions, ensuring a balanced and neutral information environment. Furthermore, simulation studies by de Arruda et al. (2022) and Wu et al. (2023) jointly explored how algorithmic adjustments could reduce polarization by promoting diverse perspectives and embedding neutral opinions. These adjustments help encourage a broader spectrum of opinions, reducing the influence of extreme opinions.

5.2. Dialogue level facilitation

Dialogue level facilitation focuses on promoting cross group communication and rational dialogue as a means of reducing polarization. These strategies aim to foster empathy and emotional connection between individuals with opposing opinions, encouraging constructive, respectful conversations that bridge ideological divides and promote mutual understanding.

Research shows that encouraging interactions based on non-political similarities can help reduce polarization. For example, a large-scale online experiment by Baliotti et al. (2021) demonstrated that matching individuals based on incidental non-political similarities could foster a sense of closeness, thereby reducing polarization. This approach encourages open engagement with opposing political opinions through structured dialogue, promoting understanding and bridging ideological divides. In a similar vein, Fernbach and Van Boven (2022) emphasized the role of structured dialogues, such as consensus conferences, in countering oversimplification and categorical thinking that drive polarization. They argued that such dialogues enable participants to engage deeply with opposing opinions, thereby reducing emotional distance and promoting cooperation. Further, an empirical study by Lee and Cho (2023) further explored how cross-cutting exposure and expression can reduce affective polarization and attitude extremity in political contexts. Their findings highlight how such dialogues help bridge ideological divides and foster more empathetic engagement with opposing opinions.

In addition to structured dialogues, other studies have examined how active nudges and participatory approaches can facilitate cross-group engagement and mitigate polarization. Judge et al. (2023) advocated for deliberative democracy and collaborative discussions to rebuild trust and address misperceptions of polarization. They argued that structured and respectful conversations allow individuals to work through their differences, fostering shared understanding and reducing polarization. Additionally, a simulation study by Sarkar and Hadfield (2024) explored how participatory institutions can encourage moderated cross group engagement by reducing the influence of extreme ideological signals.

5.3. Technology level adjustments

Technology level adjustments focus on optimizing platform algorithms and content moderation to enhance information transparency and diversity. These strategies reduce polarization by improving information flow through algorithmic interventions and network structure adjustments, thus fostering a more inclusive and balanced digital environment.

One major approach in this domain is optimizing network structures and algorithms to improve opinion dynamics. In a simulation study, Matakos et al. (2017) developed strategies like "GreedyInt" and "GreedyExt" to optimize opinion dynamics by targeting key network nodes and adjusting network topology, thereby reducing polarization. Musco et al. (2018) also used a simulation study to propose an optimization model that reduces both disagreement and polarization by finding an optimal network structure. Their model balances connections between users with similar and differing opinions, highlighting the crucial role of network design in fostering a less polarized environment. Additionally, through a modeling and simulation study, Rahaman and Hosein (2021) introduced a stochastic dynamic programming model for article recommendation systems on digital platforms. Their model aims to balance reducing polarization with user retention, showing that recommending neutral articles can help mitigate polarization in online social networks.

Further research has focused on structural adjustments to the networks themselves to counter polarization. In a modeling and simulation study, [Selleck \(2022\)](#) proposed adding a single link between stubborn nodes and their neighbors to reduce polarization in opinion networks. He used gradient-based algorithms to identify the optimal intervention for adjusting the network structure. A simulation study by [De Arruda et al. \(2022\)](#) explored the impact of social network algorithms on opinion polarization. They proposed dynamic rewiring mechanisms and post-distribution algorithms to minimize the spread of biased information and reduce the influence of echo chambers. [Einav et al. \(2022\)](#), through an empirical study, introduced a strategy for bursting filter bubbles on digital platforms. They leveraged algorithmic content curation to expose users to diverse opinions, reducing polarization and promoting open-mindedness.

In addition, strategies that modify interaction patterns within networks to limit polarization have also been explored. A modeling and simulation study by [Törnberg \(2022\)](#) suggested redesigning digital media to limit non-local interactions and enhance local cultural diversity. This approach aims to mitigate affective polarization by preventing conflicts from becoming overly partisan. In a simulation study, [Wu et al. \(2023\)](#) proposed adaptive depolarization strategies. These strategies involve building heterogeneous edges and introducing global neutral opinions into social networks, optimizing their structure to counteract polarization in both synthetic and real-world environments. A modeling and simulation study by [Racz and Rigobon \(2023\)](#) proposed a strategy for reducing polarization by perturbing social networks, specifically by adding edges between nodes with opposing opinions. Their approach leverages spectral graph theory and optimizes network structure to promote healthier interaction and reduce polarization. An empirical study by [Wang et al. \(2024\)](#) focused on how social media friend functions can reduce polarization. By enhancing normative social influence within platforms, they help mitigate the impact of extreme opinions and promote healthier discourse. Finally, a simulation study by [Sarkar and Hadfield \(2024\)](#) emphasized the role of participatory institutions in fostering moderated cross-group engagement on digital platforms. They proposed tailored content moderation and prioritizing non-ideological signals to address affective polarization.

5.4. Psychology level guidance

Psychology level guidance focuses on guiding emotional neutrality and rational expression to address the cognitive and emotional factors that fuel polarization ([Cole et al., 2025](#)). These strategies aim to adjust group identity and enhance social cohesion. By fostering empathetic and constructive interactions between opposing groups, they reduce polarization and promote understanding.

A key approach involves using psychological interventions to strengthen cognitive resistance to misinformation. Through an empirical study, [Van der Linden et al. \(2017\)](#) introduced the inoculation theory as a psychological intervention, aiming to equip individuals with cognitive resistance to misinformation. This reduces the emotional and cognitive drivers of polarization, particularly on contentious topics like climate change. In another empirical study, [Huddy and Yair \(2021\)](#) investigated how warm social relations reduce affective polarization. They found that fostering positive emotional connections between opposing groups, rather than focusing solely on policy compromise, can reduce emotional distance and promote empathy.

Additionally, several studies have focused on addressing cognitive distortions that contribute to polarization. [Fernbach and Van Boven \(2022\)](#) explored the cognitive mechanisms behind false polarization, specifically the overestimation of ideological differences. They proposed strategies to correct misperceptions, promote cognitive complexity, and encourage emotional reappraisal. An empirical study by [Lee and Cho \(2023\)](#) examined how individuals' perceptions of the social opinion climate influence their responses to opposing opinions. They found that these perceptions shape emotional reactions, which in turn affect openness to dialogue, thereby influencing cross-cutting communication.

To further illustrate how these strategies work and their effectiveness in addressing polarization, [Table 3](#) provides a summary of key strategic solutions, their characteristics, applicable scenarios, and potential limitations. By highlighting the diverse approaches within each strategy, the table reinforces the integrated nature of these efforts, providing a clear and structured summary of the literature on polarization strategic solutions. In unison, these strategies form an integrated approach to reducing polarization and fostering a more

Table 3
Analysis of multi-level strategies for addressing polarization.

Strategy Category	Characteristics	Applicable Scenarios	Potential Limitations
Information level interventions	Enhancing information diversity (Rogowski & Sutherland, 2016 ; Matakos et al., 2017) Countering misinformation (van der Linden et al., 2017 ; Judge et al., 2023)	Social networks Public policy discussions Information-sharing platforms	Information overload Resistance from entrenched biases
Dialogue level facilitation	Encouraging cross-group communication (Baliotti et al., 2021 ; Lee & Cho, 2023) Facilitating structured dialogue (Fernbach & Van Boven, 2022 ; Judge et al., 2023)	Political debates Social platform discussions Community trust rebuilding	Participant attitudes influence outcomes Extreme positions can disrupt dialogue
Technology level adjustments	Optimizing algorithmic systems (Musco et al., 2018 ; Wu et al., 2023) Reconfiguring network structures (de Arruda et al., 2022 ; Selleck, 2022)	Social media moderation Platform content curation User-generated content management	Infringing on free speech Struggling with cross-platform implementation
Psychology level guidance	Building cognitive resistance (van der Linden et al., 2017 ; Huddy & Yair, 2021) Regulating emotional responses (Fernbach & Van Boven, 2022 ; Lee & Cho, 2023)	Political polarization management Emotion-driven social issues	Requiring long-term intervention Influenced by individual differences

harmonious and understanding society.

The four strategies for addressing polarization, information level interventions, dialogue level facilitation, technology level adjustments, and psychology level guidance, work in synergy to tackle polarization from multiple perspectives. Information level interventions diversify content to reduce biases and echo chambers, while dialogue level facilitation promotes rational discourse and empathy across opposing groups. Technology level adjustments optimize platform algorithms to ensure diverse information flow, and psychology level guidance adjusts emotional and cognitive responses to reduce affective polarization. Together, these strategies provide a comprehensive approach, with each addressing polarization through a unique lens. Information strategies lay the foundation for balanced dialogue, dialogue strategies enable meaningful interactions, technology strategies enhance the infrastructure for diverse communication, and psychology strategies help overcome emotional and cognitive barriers.

6. Discussion and future directions

Building on the preceding review of models, drivers, and solutions, this chapter provides a critical synthesis by discussing the core implications, analyzing key research challenges, and outlining future directions for the field.

6.1. Discussion and implications

This section aims to develop an integrative analytical framework. By elucidating the intrinsic connections among the three core components, models, drivers, and solutions, we seek to provide a more holistic big picture for understanding the complex dynamics of polarization.

First, this study reveals a profound cross-dimensional correlation between models, drivers, and solutions. Different polarization models provide unique theoretical lenses for understanding drivers at various levels. For example, individual behavior-influenced models, by simulating mechanisms like confirmation bias and bounded rationality, precisely correspond to the psychological drivers behind polarization (Del Vicario et al., 2017; Sobkowicz, 2018). Similarly, group dynamics-driven models focus on group interaction and identity, aligning with social drivers like group identity and affective polarization (Jost et al., 2022; Kalvas et al., 2022). Meanwhile, network structure-shaped models, through analyzing network topology and algorithmic filtering, map directly onto technological drivers (Sirbu et al., 2019; de Arruda et al., 2022). This correlation reveals the inherent limitations of single-dimension interventions. For instance, merely increasing information diversity will be significantly less effective without considering the bottlenecks in information dissemination revealed by network structure models, or the deep-seated cognitive biases among psychological drivers (Bail et al., 2018). Therefore, effective solutions must be based on a comprehensive understanding of multi-dimensional models to accurately target the multiple roots of polarization.

Second, bridging empirical evidence and theoretical models is essential for understanding polarization mechanisms, with the formation of echo chambers serving as a prime illustration. At the theoretical level, numerous modeling studies, particularly bounded confidence and network evolution models incorporating algorithmic bias, have demonstrated that algorithm-driven homogeneous clustering and selective information filtering are key technological drivers of echo chamber formation (Sirbu et al., 2019; Baumann et al., 2020). At the empirical level, analyses of real-world social media data corroborate these findings, revealing that users who continuously interact within highly homogeneous online communities tend to adopt more extreme and entrenched opinions over time (Williams et al., 2015; Cinelli et al., 2021). This reciprocal validation between theoretical models and empirical evidence not only strengthens our explanatory power regarding polarization mechanisms but also allows the complex dynamics observed in real data, such as the backfire effect (Maksymov & Pogrebna, 2024), to inform model refinement and iteration, making them more realistic and nuanced.

The efficacy of group decision-making within complex social networks is fundamentally contingent upon interpersonal dynamics and the propagation of trust. However, opinion polarization poses a significant impediment that undermines the decision-making process and erodes social consensus (Proskurnikov et al., 2015; Atas et al., 2018). A profound social consequence of this polarization is the systematic restructuring of trust propagation networks (Liu et al., 2021). For social network group decision-making, these implications are systemic: it not only substantially increases the cost of achieving consensus but also exacerbates fairness concerns (Zhang et al., 2025). Furthermore, highly polarized groups become more susceptible to biased news and external guidance strategies, fostering the echo chamber effect on social media. This vulnerability not only renders the decision-making process prone to manipulation but also further entrenches the opposition between groups (Xing et al., 2022; Liu & Wen, 2024).

Finally, based on the synthesis above, the core implication of this study is that addressing opinion polarization requires cross-dimensional, synergistic intervention strategies, as single-level efforts are unlikely to be effective. An effective, integrated framework should coordinate interventions across three levels: the psychological level, through cognitive training like inoculation theory to enhance critical thinking (van der Linden et al., 2017); the social level, by facilitating structured cross-group dialogue to rebuild trust (Fernbach & Van Boven, 2022); and the technological level, by promoting algorithmic transparency and optimization to improve the information ecosystem (Musco et al., 2018). Only by organically combining these efforts can a resilient governance ecosystem be formed to address opinion polarization in the digital age.

6.2. Challenges

Despite significant advancements in understanding opinion polarization, several key challenges persist in the research landscape. These challenges stem from gaps in our comprehension of limitations in the existing models, as well as the underlying drivers and

strategic solutions. Their common root is the systemic lack of research on the complex interplay among mechanisms.

6.2.1. Limitations in opinion polarization models

A core challenge in current research is the lack of comprehensive models capable of quantifying and integrating multi-dimensional factors. Existing models often focus on a single dimension, leading to a fragmented understanding of the complex phenomenon of polarization and making it difficult to capture the interaction effects among drivers from different dimensions. This fragmentation is specifically manifested in several aspects:

First, a profound trade-off exists between the macro-level simulation of network dynamics and the micro-level characterization of individual behavior. On one hand, advanced dynamic network models, successfully simulate the evolution of network topologies over time (Yuan et al., 2022; Bolletta & Pin, 2025). However, to achieve this dynamism, they often simplify individuals into homogeneous nodes with uniform behavioral patterns. This creates a theoretical tension with the psychological drivers, as complex cognitive-motivational mechanisms are the endogenous engines of polarization (Jost et al., 2022). Conversely, models adept at depicting individual cognitive biases and emotional attachments (Sobkowicz, 2018; Quante et al., 2024) are typically based on static or simplified network assumptions.

Second, this fragmentation is reflected in the tension between models' fidelity to real-world processes and its practical utility. On one hand, in pursuit of high-fidelity simulations of the real world, researchers have developed complex ABMs that can finely reproduce psychological processes like cognitive filtering and emotional contagion. However, the numerous parameters and emergent behaviors of these models often render them black boxes that are difficult to analyze, thereby limiting their interpretability and practical utility in guiding specific policy interventions. On the other hand, more parsimonious mathematical models are highly interpretable, but their modeling of biases and emotions is often oversimplified into a fixed stubbornness parameter (Baldassarri & Page, 2021), which weakens their explanatory power in real-world scenarios. Striking a balance between model complexity and interpretability to build models that are both realistic and usable remains a significant challenge.

Finally, the exploration of artificial intelligence's (AI) role in the polarization ecosystem severely lags, highlighting the inadequacy of integrating technological and psychological dimensions. Existing models typically simplify AI algorithms into a static, homogeneity-enhancing technological driver. However, the essence of modern AI is its adaptivity. It forms a co-evolutionary feedback loop with user behavior: the algorithm adjusts its strategy in real-time based on user interactions, while user preferences are, in turn, shaped and solidified by the algorithm. Current models fail to capture how AI, as an adaptive agent capable of continuous learning and adaptation, rather than merely a static structure, amplifies and locks in polarized states (Govers et al., 2024; Xing et al., 2024). This leaves our ability to predict and evaluate interventions for AI-driven polarization phenomena severely underdeveloped (Xie et al., 2025).

6.2.2. Research gaps in the drivers of opinion polarization

First, existing research has largely overlooked the critical moderating role of individual heterogeneity. Most models assume that individual behavior within networks is homogeneous (Bail et al., 2018; Fraiman et al., 2025), an assumption that is far from reality. A critical consequence of this homogeneity assumption is that models cannot explain why, in the same network environment, different individuals can have diametrically opposed reactions. An individual with high cognitive flexibility will react very differently to opposing information than an individual driven by strong identity-protective motivations. By failing to effectively integrate these heterogeneity variables, which are rooted in psychological drivers, the ability of existing models to explain key phenomena like the backfire effect is significantly diminished.

Second, the systematic neglect of platform differences is another significant shortcoming. Current research often treats polarization as a universal process (Liu et al., 2023b; Valensise et al., 2023), overlooking the vast differences among social media platforms (such as X, Facebook, and Reddit). These platforms vary significantly in terms of network structure, interaction norms, and information diffusion mechanisms. This oversight is problematic because it obscures the important possibility that different platforms may require entirely different modeling paradigms. Current research generally lacks the comparative analysis needed to match platform characteristics with appropriate model selections. This absence severely limits our understanding of polarization mechanisms in different digital environments. It also hinders our ability to design effective, platform-specific intervention strategies.

6.2.3. Shortcomings for strategic solutions to address opinion polarization

Although strategic solutions have been proposed to address polarization, their effectiveness in practice is often limited. The fundamental reason lies in the profound conflict between the single-dimension nature of these solutions and the systemic nature of polarization's drivers.

First, the effects of technical interventions at the information or algorithmic level are often nullified by powerful psychological and network drivers. For instance, recommending diverse information to users via algorithms aims to break echo chambers (Hazla et al., 2024), but this approach directly conflicts with deep-seated psychological drivers like confirmation bias and the backfire effect. As a result, users may become more resistant to opposing opinions, paradoxically intensifying polarization. Furthermore, network structure models also predict the limitations of such interventions: in highly homophilous online communities, even if information sources are diverse, the information itself may struggle to effectively cross community boundaries. Therefore, these seemingly direct solutions often fail to address the root cause, as they do not adequately account for the complexities at the psychological and network levels.

Second, social-level interventions aimed at promoting cross-group communication often confuse the types of polarization, leading to a mis-targeting of the problem. Many dialogue strategies are designed to resolve disagreements on issues, but the primary obstacle today is often the emotional animosity and distrust between groups (Shi et al., 2019). When groups are already fraught with antagonistic emotions, merely creating opportunities for dialogue without first addressing the emotional divide is often ineffective. As long

as the foundational problem of affective polarization remains untouched, any effort to promote rational dialogue is likely to remain superficial due to a lack of trust and goodwill.

Finally, even with the recognition that single-dimension interventions are insufficient, synergistic intervention strategies face immense integration and evaluation challenges in practice. The vast majority of existing studies test single-level interventions in isolated contexts, leaving little known about the synergistic effects or potential negative interactions of multi-dimensional approaches. Moreover, implementing psychological or social-level interventions on large-scale platforms faces significant ethical and feasibility barriers. This creates a vast chasm between the theoretical advocacy for synergistic governance and its effective implementation in practice.

6.2.4. *Lack of integrative research on mechanism interplay*

Although pioneering studies on opinion polarization, as highlighted in previous sections, have begun to integrate multiple mechanisms (Caluwaerts et al., 2023), the field as a whole has yet to systematically adopt this integrative approach. A significant challenge, therefore, is the lack of research that systematically models and empirically tests these complex interactions of opinion polarization (Biondi et al., 2023). For instance, critical questions remain unexplored: how does the impact of a specific cognitive bias change under different algorithmic filtering regimes? Or how might a given network structure amplify or dampen the feedback loop between emotional expression and technological amplification? This prevalent lack of an integrative focus is a key reason why many current models struggle to capture the complex, non-linear dynamics of real-world polarization and why single-domain interventions often have limited effects.

6.3. *Future directions*

To address the challenges in existing research and advance the field, future studies should focus on the following four interconnected directions. These directions aim to enhance the comprehensiveness and predictive power of models, deepen the understanding of polarization's drivers, develop more effective intervention strategies, and foster interdisciplinary exploration of the interplay among mechanisms.

6.3.1. *Advancing opinion polarization models*

To overcome the limitations of fragmentation and oversimplification in current models, future model development should focus on creating hybrid dynamic models. This entails integrating the nuanced depiction of rich psychological processes from ABMs, such as cognitive biases and emotional attachments, with dynamic frameworks capable of simulating large-scale network topology evolution. Furthermore, models must treat AI as an adaptive agent that co-evolves with user behavior to more realistically capture its complex role in the polarization ecosystem. The ultimate goal is to construct new models that possess both macro-level predictive power and micro-level explanatory ability, while balancing real-world complexity with practical utility.

6.3.2. *Addressing research gaps in the drivers of opinion polarization*

To address the current neglect of individual and platform differences, future work must vigorously promote empirical research that considers contextual factors. At the individual level, research should shift from analyzing average effects to exploring heterogeneous effects. Through experiments or data analysis, studies should examine how different psychological traits moderate individual responses in various information environments. This will deepen our understanding of complex phenomena like the backfire effect. At the platform level, systematic cross-platform comparative analyses should be conducted to investigate how the unique structures and norms of different platforms (e.g., X, Facebook, and Reddit) foster different types and speeds of polarization. Such research should also explore which theoretical models are best suited to explain the phenomena on specific platforms, thereby providing a solid evidence base for developing platform-specific intervention strategies.

6.3.3. *Developing more effective strategic solutions to address opinion polarization*

Given the fundamental shortcomings of single-dimension interventions, future research on solutions must shift toward designing and evaluating multi-level, synergistic strategies. Researchers should explore how to organically combine technical-level algorithmic optimizations (such as enhancing the diversity and fairness of recommended content), social-level structured dialogue (such as facilitating moderate cross-group interactions), and psychological-level cognitive guidance (such as inoculation-based media literacy education). More critically, future research must directly confront the ethical, feasibility, and scalability challenges that synergistic interventions face in practice. For example, how can psychological interventions be implemented ethically on large-scale social platforms? How can the synergistic or conflicting effects of different interventions be quantitatively evaluated? Answering these questions is key to transforming the ideal framework of synergistic governance into a practical and usable toolkit.

6.3.4. *Fostering interdisciplinary research on mechanism interplay*

A crucial future direction is to make the study of how polarization mechanisms interact a central focus of research. This will require interdisciplinary teams to systematically investigate the interplay between drivers from different dimensions. For instance, they could test how a specific psychological driver performs under various technological conditions, or how a given network structure amplifies or dampens social feedback loops. To achieve this, researchers should work on developing co-evolutionary models that can track how individual opinions, social connections, and platform algorithms influence one another over time. Furthermore, large-scale experiments comparing different platforms are needed to understand how their unique designs affect this interplay. This kind of research

naturally requires collaboration across different fields: computer scientists build the models, social psychologists provide realistic rules for human behavior, and sociologists explain the broader social context. Only through such deep collaboration can the field develop a truly comprehensive understanding of polarization dynamics.

7. Conclusions

In this paper, we have reviewed the phenomenon of opinion polarization and its underlying drivers, and introduced a conceptual framework for exploring polarization. We then examined several of the most widely studied models of polarization, including individual behavior-influenced polarization models, group dynamics-driven polarization models, and network structure-shaped polarization models. Additionally, we reviewed the drivers of polarization, highlighting the psychological, social, and technological factors that contribute to its dynamics. We also review various strategic solutions, focusing on information level interventions, dialogue level facilitation, technology level adjustments, and psychology level guidance. Finally, several key challenges in the existing research on opinion polarization were identified, including the reliance on static network structures, insufficient consideration of cognitive biases, and the limited exploration of artificial intelligence's impact on polarization. Additionally, current models often fail to capture the dynamic, nonlinear nature of polarization and lack interpretability when applied to real-world scenarios. To address these gaps, we propose several future directions. First, polarization models should evolve to incorporate dynamic network structures and individual behavior changes over time, improving their predictive accuracy and real-world applicability. Second, future research should examine the diverse drivers of polarization, with particular attention to network heterogeneity and platform differences. Third, more effective strategic solutions should be developed, focusing not only on algorithmic interventions but also on fostering cross-group dialogue and reducing antagonism. Finally, AI-driven tools and their impact on polarization should be prioritized, with a focus on enhancing algorithmic transparency and understanding their broader implications for polarization. Advancing these areas will deepen our understanding of opinion polarization in contemporary society and inform more effective strategic solutions.

CRedit authorship contribution statement

Shenghua Liu: Writing – original draft, Methodology, Investigation, Formal analysis, Conceptualization. **Zhibin Wu:** Writing – original draft, Supervision, Methodology, Investigation, Funding acquisition, Conceptualization. **Luis Martínez:** Writing – review & editing, Supervision, Methodology, Investigation, Conceptualization.

Acknowledgments

This work was supported by the National Natural Science Foundation of China under Grants 72371175 and 71971148.

Appendix A

The following Table A1 provides key mathematical models. Each model is categorized based on its focus, and the mathematical formulations, core assumptions, and key references are provided.

Table A1
Mathematical models of opinion polarization.

Model	General form	Model Description
Bounded confidence models (BC model)	$x_i(t+1) = \begin{cases} x_i(t) + \mu(x_j - x_i), & x_i - x_j < \varepsilon \\ x_i(t) - \mu(x_j - x_i), & x_i - x_j \geq \varepsilon \end{cases}$ $o_i(t+1) = o_i(t) + \alpha \cdot \frac{1}{ N_i(t) } \sum_{j \in N_i(t)} (o_j(t) - o_i(t))$ $\frac{dX_i^t}{dt} = -\alpha_i X_i^t + \alpha_i \sum_{j \in \Omega} g_i(\Omega_i, \Omega_j) f_i(X_i^t, X_j^t, \varepsilon_i) X_j^t$	Rewire with unbounded confidence model Balance opinion updating and network rewiring (Del Vicario et al., 2017) Hegselmann-Krause dynamics with conformity Identity-constrained interaction (Kalvas et al., 2022) Multi-population BC model Balance scalability and accuracy (Bakaryan et al., 2025)

(continued on next page)

Table A1 (continued)

Model	General form	Model Description
DeGroot models	$x_i(t+1) = \begin{cases} \theta_i x_i(t) + (1-\theta_i) \sum_{j \in N_i} q_{ij} x_j(t), & \text{if } t \in \{s_i\} \\ x_i(t), & \text{otherwise} \end{cases}$ $x_i(t)$: Opinion of agent i at time t θ_i : Self-confidence level of agent i N_i : Neighbor set of agent i q_{ij} : Standardized weight $R_i(k+1) = \sum_{r=1}^{d_i} c(i,r) R_{i(l(r))}(k) + W_i(k) + (1-c-d) R_i(k)$ $R_i(k)$: Opinion of individual i at time k $c(i,r)$: Weight placed by individual i on their r -th inbound neighbor $W_i(k)$: External signal received by individual i at time k c and d : Parameters representing trust in neighbors and media signals	Quasi-Structural balance Asynchronous opinion updates (Shi et al., (2023)) Influenced by both neighbors and biased media signals Selective exposure (Fraiman et al., 2025)
Friedkin-Johnsen models	$x_i(k+1) = \lambda_{ii} C \sum_{j=1}^n w_{ij} x_j(k) + (1-\lambda_{ii}) u_i$ $x_i(k)$: Opinion of agent i at time step k λ_{ii} : Susceptibility of agent i to social influence C : Multi-issue dependence structure matrix w_{ij} : Influence weight from agent j to agent i u_i : Initial opinion or prejudice of agent i $x_i = \frac{b_i + \sum_j F_{ij} x_j}{1 + \sum_j F_{ij}}$ x_i : Expressed opinion of node i at a given time b_i : Innate opinion of node i F_{ij} : Weight of the edge between node i and j	Multidimensional opinion dynamics Multi-issue dependence structure (Parsegov et al., 2017) Co-evolving dynamics Confirmation bias integration Friend-of-Friend mechanism (Bhalla et al., 2023)
Geometric models of opinion dynamics	$X_{t+1}^{(i)} \propto X_t^{(i)} + \eta \cdot \text{sgn}(\langle X_t^{(i)}, \xi_t \rangle) \xi_t$ $X_t^{(i)}$: Opinion of agent i at time t ξ_t : A random vector $\langle X_t^{(i)}, \xi_t \rangle$: The inner product η : A fixed step size $u_i^{(t+1)} = \frac{w_i^{(t+1)}}{\ w_i^{(t+1)}\ }$ $w_i^{(t+1)} = u_i^{(t)} + \eta \cdot \langle u_i^{(t)}, v^{(t)} \rangle v^{(t)}$ $u_i^{(t)}$: Opinion of agent i at time t $v^{(t)}$: Intervention vector at time t η : A global parameter $\langle u_i^{(t)}, v^{(t)} \rangle$: Inner product	Signed HJMR model Strong polarization Binary decision dynamics (Gaitonde et al., 2021) Multidimensional opinion space Explicit external interventions (Hazla et al., 2024)
Agent-based models	$X_j(\theta, t+1) = [X_j(\theta, t) FL(S_i, \theta)]_N$ $X_j(\theta, t)$: Belief distribution of agent j over topic θ at time t $FL(S_i, \theta)$: Filtered likelihood of the information source S_i regarding θ $[\cdot]_N$: Normalization operator $\frac{\Delta A_i}{\Delta t} = \frac{ N_i(t) }{\tau} [(1-\gamma_i) \cdot \text{sign}(N_i(t)) + \gamma_i (A_i^* - A_i(t-1))]$ A_i^* : Initial conviction of agent i γ_i : Self-reliance parameter of agent i $A_i(t)$: Current opinion of agent i at time t $N_i(t)$: Neighbourhood influence on agent i τ : Time scale parameter	Cognitive bias filtering Bayesian like framework Filtering mechanism (Sobkowicz, 2018) Incorporate self-reliance Balances individual conviction and neighborhood influence (Quante et al., 2024)
Social reinforcement learning models	$Q_i(o) \leftarrow \begin{cases} (1-\alpha) Q_i(o) + \alpha r_i & \text{if } o = \text{expression} \\ Q_i(o) & \text{else} \end{cases}$ $Q_i(o)$: Internal evaluation of opinion o by agent i α : Learning rate. r_i : Reward received by agent i $Q_i(o, t+1) = (1-\alpha) Q_i(o, t) + \alpha R_j$ $Q_i(o, t)$: Agent i 's quality estimate for opinion o at time t α : Learning rate R_j : Reward or punishment from neighbor j	Reinforcement learning Social feedback Exploration-driven diversity (Banisch & Olbrich, 2019) Reinforcement learning dynamics Opinion quality updates Feedback-driven polarization (Meylahn & Meylahn, 2024)

To prevent ambiguity, Table A2 highlights the key distinctions between overlapping terms.

Table A2
Clarification of overlapping term pairs.

Term Pair	Key Distinction
Opinion polarization vs. group polarization vs. Bi-polarization	The overarching phenomenon of opinion divergence and division vs. group discussion reinforces and extremizes initial attitudes vs. opinion distribution splits into two opposing, extreme camps
Selective communication vs. Selective interaction	Focus on the content of exchange (what is said) vs. the choice of partners (who to talk to)
Group dynamics vs. Group interaction dynamics	The broader field of group behavior vs. specific models of direct exchange within groups
Opinion amplification vs. Emotional amplification	platforms/technologies disproportionately increase the visibility of certain opinions vs. emotional reactions intensify hostility and polarization
Network fragmentation vs. Community structure	The network's subgroup arrangement vs. the outcome of disconnection between those subgroups
Dynamic interactions vs. Dynamic network models	Opinion evolution on a static network vs. evolution of the network structure itself

Data availability

No data was used for the research described in the article.

References

- AlMahmoud, H., & Mallmann-Trenn, F. (2024). Harmonizing vs polarizing platform influence functions. In , 1190. *Proceedings of the international conference on complex networks and their applications* (pp. 152–165). Springer.
- Arendt, D. L., & Blaha, L. M. (2015). Opinions, influence, and zealotry: A computational study on stubbornness. *Computational and Mathematical Organization Theory*, 21(2), 184–209.
- Arora, S. D., Singh, G. P., Chakraborty, A., & Maity, M. (2022). Polarization and social media: A systematic review and research agenda. *Technological Forecasting and Social Change*, 183, Article 121942.
- Atas, M., Reiterer, S., Felfernig, A., Tran, T. N. T., & Stettinger, M. (2018). Polarization effects in group decisions. In *Proceedings of the adjunct publication of the 26th conference on user modeling, adaptation and personalization* (pp. 305–310).
- Axelrod, R. (1997). The dissemination of culture-A model with local convergence and global polarization. *Journal of Conflict Resolution*, 41(2), 203–226.
- Azzopardi, L. (2021). Cognitive biases in search: A review and reflection of cognitive biases in information retrieval. In *Proceedings of the conference on human information interaction and retrieval* (pp. 27–37).
- Bail, C. A., Argyle, L. P., Brown, T. W., Bumpus, J. P., Chen, H. H., Hunzaker, M. B. F., Lee, J., Mann, M., Merhout, F., & Volfovsky, A. (2018). Exposure to opposing views on social media can increase political polarization. *Proceedings of the National Academy of Sciences of the United States of America*, 115(37), 9216–9221.
- Bakaryan, T., Gu, Y. L., Hovakimyan, N., Abdelzaher, T., & Lebiere, C. (2025). Multi-population opinion dynamics model. *Nonlinear Dynamics*, 113(2), 1925–1952.
- Baldassarri, D., & Page, S. E. (2021). The emergence and perils of polarization. *Proceedings of the National Academy of Sciences of the United States of America*, 118(50), Article e2116863118.
- Balietti, S., Getoor, L., Goldstein, D. G., & Watts, D. J. (2021). Reducing opinion polarization: effects of exposure to similar people with differing political views. *Proceedings of the National Academy of Sciences of the United States of America*, 118(52), Article e2112552118.
- Banisch, S., & Olbrich, E. (2019). Opinion polarization by learning from social feedback. *Journal of Mathematical Sociology*, 43(2), 76–103.
- Banisch, S., & Shamon, H. (2025). Biased processing and opinion polarization: Experimental refinement of argument communication theory in the context of the energy debate. *Sociological Methods & Research*, 54(1), 187–236.
- Barberá, P. (2014). How social media reduces mass political polarization. Evidence from Germany, Spain, and the US. *Job Market Paper, New York University*, 46, 1–46.
- Baron, J. W. (2021). Consensus, polarization, and coexistence in a continuous opinion dynamics model with quenched disorder. *Physical Review E*, 104(4), Article 044309.
- Baumann, F., Lorenz-Spreen, P., Sokolov, I. M., & Starnini, M. (2020). Modeling echo chambers and polarization dynamics in social networks. *Physical Review Letters*, 124(4), Article 048301.
- Baumann, F., Lorenz-Spreen, P., Sokolov, I. M., & Starnini, M. (2021). Emergence of polarized ideological opinions in multidimensional topic spaces. *Physical Review X*, 11(1), Article 011012.
- Berman, R., & Katona, Z. (2020). Curation algorithms and filter bubbles in social networks. *Marketing Science*, 39(2), 296–316.
- Bessi, A., Petroni, F., Del Vicario, M., Zollo, F., Anagnostopoulos, A., Scala, A., Caldarelli, G., & Quattrociocchi, W. (2016). Homophily and polarization in the age of misinformation. *European Physical Journal-Special Topics*, 225(10), 2047–2059.
- Bhalla, N., Lechowicz, A., & Musco, C. (2023). Local edge dynamics and opinion polarization. In *Proceedings of the sixteenth ACM international conference on web search and data mining* (pp. 6–14).
- Biondi, E., Boldrini, C., Passarella, A., & Conti, M. (2023). Dynamics of opinion polarization. *IEEE Transactions on Systems Man Cybernetics-Systems*, 53(9), 5381–5392.
- Bliuc, A. M., Bettis, J. M., Vergani, M., Bouguettaya, A., & Cristea, M. (2024). A theoretical framework for polarization as the gradual fragmentation of a divided society. *Communications Psychology*, 2(1), 75.
- Bliuc, A. M., Bouguettaya, A., & Felise, K. D. (2021). Online intergroup polarization across political fault lines: An integrative review. *Frontiers in Psychology*, 12, Article 641215.
- Bolletta, U., & Pin, P. (2025). Dynamic opinion updating with endogenous networks. *European Economic Review*, 176, Article 105045.
- Borges, H. M., Vasconcelos, V. V., & Pinheiro, F. L. (2024). How social rewiring preferences bridge polarized communities. *Chaos Solitons & Fractals*, 180, Article 114594.
- Brooks, H. Z., Chodrow, P. S., & Porter, M. A. (2024). Emergence of polarization in a sigmoidal bounded-confidence model of opinion dynamics. *SIAM Journal on Applied Dynamical Systems*, 23(2), 1442–1470.
- Cai, M., Luo, H., Meng, X., Cui, Y., & Wang, W. (2023). Network distribution and sentiment interaction: Information diffusion mechanisms between social bots and human users on social media. *Information Processing and Management*, 60(2), Article 103197.
- Caluwaerts, D., Bernaerts, K., Kesberg, R., Smets, L., & Spruyt, B. (2023). Deliberation and polarization: A multi-disciplinary review. *Frontiers in Political Science*, 5, Article 1127372.
- Cao, F., & Reed, S. (2025). The iterative persuasion-polarization opinion dynamics and its mean-field analysis. *SIAM Journal on Applied Mathematics*, 85(4), 1596–1620.

- Chen, T. G., Li, Q. Q., Fu, P. H., Yang, J. J., Xu, C. H., Cong, G. D., & Li, G. F. (2020). Public opinion polarization by individual revenue from the social preference theory. *International Journal of Environmental Research and Public Health*, 17(3), 946.
- Cinelli, M., Morales, G. D., Galeazzi, A., Quattrociocchi, W., & Starnini, M. (2021). The echo chamber effect on social media. *Proceedings of the National Academy of Sciences of the United States of America*, 118(9), Article e2023301118.
- Cisneros-Velarde, P., Chan, K. S., & Bullo, F. (2021). Polarization and fluctuations in signed social networks. *IEEE Transactions on Automatic Control*, 66(8), 3789–3793.
- Cole, J. C., Gillis, A. J., van der Linden, S., Cohen, M. A., & Vandenberg, M. P. (2025). Social psychological perspectives on political polarization: Insights and implications for climate change. *Perspectives on Psychological Science*, 20(1), 115–141.
- Cyrancka, J., & Mucha, P. B. (2024). A polarization opinion model inspired by bounded confidence communications. *Journal of the Franklin Institute-Engineering and Applied Mathematics*, 361(16), Article 107140.
- Dai, J. L., Zhu, J. M., & Wang, G. Q. (2022). Opinion influence maximization problem in online social networks based on group polarization effect. *Information Sciences*, 609, 195–214.
- Dalton, R. J. (2006). Social modernization and the end of ideology debate: Patterns of ideological polarization. *Japanese Journal of Political Science*, 7(1), 1–22.
- Dalton, R. J. (2021). Modeling ideological polarization in democratic party systems. *Electoral Studies*, 72, Article 102346.
- Dandekar, P., Goel, A., & Lee, D. T. (2013). Biased assimilation, homophily, and the dynamics of polarization. *Proceedings of the National Academy of Sciences of the United States of America*, 110(15), 5791–5796.
- Das, S., Dutta, S., Chakraborty, S., & Biswas, S. (2023). Influence of opinion formation on polarization in social networks. In *Proceedings of the 14th international conference on computing communication and networking technologies* (pp. 1–7).
- Das, A., Gollapudi, S., & Munagala, K. (2014). Modeling opinion dynamics in social networks. In *Proceedings of the 7th ACM international conference on web search and data mining* (pp. 403–412). New York, NY: Assoc Computing Machinery.
- de Arruda, H. F. R., Cardoso, F. M., de Arruda, G. F., Hernandez, A. R., Costa, L. D., & Moreno, Y. (2022). Modelling how social network algorithms can influence opinion polarization. *Information Sciences*, 588, 265–278.
- De Marzo, G., Zaccaria, A., & Castellano, C. (2020). Emergence of polarization in a voter model with personalized information. *Physical Review Research*, 2(4), Article 043117.
- DeGroot, M. H. (1974). Reaching a consensus. *Journal of the American Statistical Association*, 69(345), 118–121.
- Del Vicario, M., Scala, A., Caldarelli, G., Stanley, H. E., & Quattrociocchi, W. (2017). Modeling confirmation bias and polarization. *Scientific Reports*, 7, Article 40391.
- Dimant, E. (2024). Hate trumps love: the impact of political polarization on social preferences. *Management Science*, 70(1), 1–31.
- Druckman, J. N., & Levendusky, M. S. (2019). What do we measure when we measure affective polarization? *Public Opinion Quarterly*, 83(1), 114–122.
- Drummond, C., & Fischhoff, B. (2017). Individuals with greater science literacy and education have more polarized beliefs on controversial science topics. *Proceedings of the National Academy of Sciences of the United States of America*, 114(36), 9587–9592.
- Du, S. Y., & Gregory, S. (2016). The echo chamber effect in Twitter: Does community polarization increase?. In *Proceedings of the 5th international workshop on complex networks and their applications* (pp. 373–384). Univ Milan, Milan, ITALY: Springer International Publishing Ag.
- Einav, G., Allen, O., Gur, T., Maaravi, Y., & Ravner, D. (2022). Bursting filter bubbles in a digital age: Opening minds and reducing opinion polarization through digital platforms. *Technology in Society*, 71, Article 102136.
- Feezell, J. T., Wagner, J. K., & Conroy, M. (2021). Exploring the effects of algorithm-driven news sources on political behavior and polarization. *Computers in Human Behavior*, 116, Article 106626.
- Feliciani, T., Flache, A., & Tolsma, J. (2017). How, when and where can spatial segregation induce opinion polarization? Two competing models. *The Journal of Artificial Societies and Social Simulation*, 20(2), 6.
- Fernbach, P. M., & Van Boven, L. (2022). False polarization: Cognitive mechanisms and potential solutions. *Current Opinion in Psychology*, 43, 1–6.
- Flache, A., Mäs, M., Feliciani, T., Chattoe-Brown, E., Deffuant, G., Huet, S., & Lorenz, J. (2017). Models of social influence: Towards the next frontiers. *The Journal of Artificial Societies and Social Simulation*, 20(4), 2.
- Fraiman, N., Lin, T. C., & Olvera-Cravioto, M. (2025). Opinion dynamics on directed complex networks. *Mathematics of Operations Research*, 50(2), 838–870.
- Franceschi, J., Pareschi, L., Bellodi, E., Gavanelli, M., & Bresadola, M. (2023). Modeling opinion polarization on social media: Application to Covid-19 vaccination hesitancy in Italy. *Plos One*, 18(10), Article e0291993.
- French, J. R., Jr (1956). A formal theory of social power. *Psychological Review*, 63(3), 181–194.
- Gaitonde, J., Kleinberg, J., & Tardos, E. (2021). Polarization in geometric opinion dynamics. In *Proceedings of the 22nd ACM conference on economics and computation* (pp. 499–519).
- Galam, S. (2023). Unanimity, coexistence, and rigidity: three sides of polarization. *Entropy*, 25(4), 622.
- García-Zamora, D., Labella, Á., Rodríguez, R. M., & Martínez, L. (2021). Nonlinear preferences in group decision-making. Extreme values amplifications and extreme values reductions. *International Journal of Intelligent Systems*, 36(11), 6581–6612.
- Ghenai, A., Noorian, Z., Moradizadeh, H., Abadeh, P., Erentzen, C., & Zarrinkalam, F. (2025). Exploring hate speech dynamics: The emotional, linguistic, and thematic impact on social media users. *Information Processing and Management*, 62(3), Article 104079.
- Govers, J., Velloso, E., Kostakos, V., & Goncalves, J. (2024). AI-driven mediation strategies for audience depolarisation in online debates. In *Proceedings of the CHI conference on human factors in computing systems* (pp. 1–18).
- Gubanov, D., & Petrov, I. (2019). Multidimensional model of opinion polarization in social networks. In *Proceedings of the twelfth international conference "management of large-scale system development"* (pp. 1–4).
- Gubanov, D. A., Petrov, I. V., & Chkhartishvili, A. G. (2021). Multidimensional model of opinion dynamics in social networks: Polarization indices. *Automation and Remote Control*, 82(10), 1802–1811.
- Guilbeault, D., Van Loon, A., Lix, K., Goldberg, A., & Srivastava, S. B. (2023). Exposure to the views of opposing others with latent cognitive differences results in social influence-but only when those differences remain obscured. *Management Science*, 70(10), 6669–6684.
- Guo, J., & Hu, Y. (2023). Does social media use polarize or depolarize political opinion in China? Explaining opinion polarization within an extended communication mediation model. *Social Media + Society*, 9(3), Article 20563051231196899.
- Harel, T. O., Maoz, I., & Halperin, E. (2020). A conflict within a conflict: Intragroup ideological polarization and intergroup intractable conflict. *Current Opinion in Behavioral Sciences*, 34, 52–57.
- Hazla, J., Jin, Y., Mossel, E., & Ramnarayan, G. (2024). A geometric model of opinion polarization. *Mathematics of Operations Research*, 49(1), 251–277.
- Huddy, L., & Yair, O. (2021). Reducing affective polarization: Warm group relations or policy compromise? *Political Psychology*, 42(2), 291–309.
- Iyengar, S., Lelkes, Y., Levendusky, M., Malhotra, N., & Westwood, S. J. (2019). The origins and consequences of affective polarization in the United States. *Annual Review of Political Science*, 22(1), 129–146.
- Iyengar, S., & Westwood, S. J. (2015). Fear and loathing across party lines: New evidence on group polarization. *American Journal of Political Science*, 59(3), 690–707.
- Jost, J. T., Baldassarri, D. S., & Druckman, J. N. (2022). Cognitive-motivational mechanisms of political polarization in social-communicative contexts. *Nature Reviews Psychology*, 1(10), 560–576.
- Judge, M., Kashima, Y., Steg, L., & Dietz, T. (2023). Environmental decision-making in times of polarization. *Annual Review of Environment and Resources*, 48, 477–503.
- Kadiyala, B., & Shin, D. (2024). Social learning and content quality under polarization. *Manufacturing & Service Operations Management*, 26(6), 2237–2255.
- Kalvas, F., Ramaswamy, A., & Slater, M. D. (2022). Identity drives polarization: Advancing the Hegselmann-Krause model by identity groups. In *Proceedings of the conference of the European social simulation association* (pp. 249–262).
- Keijzer, M. A., Mäs, M., & Flache, A. (2024). Polarization on social media: Micro-level evidence and macro-level implications. *The Journal of Artificial Societies and Social Simulation*, 27(1), 7.
- Kim, J., & Rojas, H. (2025). How does political communication relate to perceived extremity of partisans and their divides? From communication to polarization perceptions to political engagement. *Communication Studies*, 76(1), 48–71.
- Kish Bar-On, K., Dimant, E., Lelkes, Y., & Rand, D. G. (2024). Unraveling polarization: Insights into individual and collective dynamics. *PNAS Nexus*, 3(10), 426.

- Kreiss, D., & McGregor, S. C. (2024). A review and provocation: On polarization and platforms. *New Media & Society*, 26(1), 556–579.
- Krishnaro, S., & Selvamani, S. A. (2024). Impact of the network size and frequency of information receipt on polarization in social networks. *Complexity*, 2024, Article 4742401.
- Kwon, H. E., Oh, W., & Kim, T. (2017). Platform structures, homing preferences, and homophilous propensities in online social networks. *Journal of Management Information Systems*, 34(3), 768–802.
- Lechiakh, M., El-Moutaouakkil, Z., & Maurer, A. (2024). Towards long-term depolarized interactive recommendations. *Information Processing and Management*, 61(6), Article 103833.
- Lee, S., & Cho, J. H. (2023). Hearing and speaking the other side: The roles of expression and opinion climate perception in political polarization. *Computers in Human Behavior*, 143, Article 107672.
- Lees, J., & Cikara, M. (2021). Understanding and combating misperceived polarization. *Philosophical Transactions of the Royal Society B*, 376(1822), Article 20200143.
- Levendusky, M. S. (2010). Clearer cues, more consistent voters: A benefit of elite polarization. *Political Behavior*, 32(1), 111–131.
- Levin, S. A., Milner, H. V., & Perrings, C. (2021). The dynamics of political polarization. *Proceedings of the National Academy of Sciences of the United States of America*, 118(50), Article e2116950118.
- Levy, R. (2021). Social media, news consumption, and polarization: Evidence from a field experiment. *American Economic Review*, 111(3), 831–870.
- Li, Q., Li, Y. L., Zhang, S. H., Zhou, X., & Pan, Z. Y. (2025). A theoretical framework for human-centered intelligent information services: A systematic review. *Information Processing and Management*, 62(1), Article 103891.
- Li, L., Scaglione, A., Swami, A., & Zhao, Q. (2013). Consensus, polarization and clustering of opinions in social networks. *IEEE Journal on Selected Areas in Communications*, 31(6), 1072–1083.
- Li, Z. P., & Tang, X. J. (2023). Structure balance and opinions dynamic in signed social network. *Journal of Systems Science & Complexity*, 36(4), 1626–1640.
- Lim, S. L., & Bentley, P. J. (2022). Opinion amplification causes extreme polarization in social networks. *Scientific Reports*, 12(1), Article 18131.
- Liu, S., Guan, X. W., & Xia, H. X. (2023a). Exploring group opinion polarization based on an opinion dynamics model considering positive and negative social influence. In *Proceedings of the international symposium on knowledge and systems sciences* (pp. 260–273).
- Liu, J. Z., Huang, S. D., Aden, N. M., Johnson, N. F., & Song, C. M. (2023b). Emergence of polarization in coevolving networks. *Physical Review Letters*, 130(3), Article 037401.
- Liu, S., & Wen, H. (2024). Agent-based modelling of polarized news and opinion dynamics in social networks: A guidance-oriented approach. *Journal of Complex Networks*, 12(4), cnae028.
- Liu, Z., Deng, Y., & Yager, R. R. (2021). Measure-based group decision-making with principle-guided social interaction influence for incomplete information: A game theoretic perspective. *IEEE Transactions on Fuzzy Systems*, 30(4), 1149–1163.
- Loy, N., Raviola, M., & Tosin, A. (2022). Opinion polarization in social networks. *Philosophical Transactions of the Royal Society A*, 380(2224), Article 20210158.
- Luo, S. L., Xia, H. X., & Yin, B. R. (2014). Continuous opinion dynamics on an adaptive coupled random network. *Advances in Complex Systems*, 17(03n04), Article 1450012.
- Macias, R. C., & Vera, J. M. R. (2024). Dynamics of opinion polarization in a population. *Mathematical Social Sciences*, 128, 31–40.
- Makymov, I. S., & Pogrebná, G. (2024). Quantum-mechanical modelling of asymmetric opinion polarisation in social networks. *Information*, 15(3), 170.
- Mäs, M., & Flache, A. (2013). Differentiation without distancing. Explaining bi-polarization of opinions without negative influence. *Plos One*, 8(11), Article e74516.
- Mason, L. (2013). The rise of uncivil agreement: Issue versus behavioral polarization in the American electorate. *American Behavioral Scientist*, 57(1), 140–159.
- Matakos, A., Terzi, E., & Tsaparas, P. (2017). Measuring and moderating opinion polarization in social networks. *Data Mining and Knowledge Discovery*, 31(5), 1480–1505.
- Meylahn, B. V., & Meylahn, J. M. (2024). How social reinforcement learning can lead to metastable polarisation and the voter model. *Plos One*, 19(12), Article e0313951.
- Mobilis, M. (2023). Polarization and consensus in a voter model under time-fluctuating influences. *Physics*, 5(2), 517–536.
- Moscovici, S., & Zavalloni, M. (1969). The group as a polarizer of attitudes. *Journal of Personality Social Psychology*, 12(2), 125–135.
- Mueller, S. T., & Tan, Y. Y. S. (2018). Cognitive perspectives on opinion dynamics: The role of knowledge in consensus formation, opinion divergence, and group polarization. *Journal of Computational Social Science*, 1, 15–48.
- Musco, C., Musco, C., & Tsourakakis, C. E. (2018). Minimizing polarization and disagreement in social networks. In *Proceedings of the world wide web conference* (pp. 369–378).
- Mutz, D. C. (2006). *Hearing the other side: Deliberative versus participatory democracy*. Cambridge University Press.
- O'Connor, C., & Weatherall, J. O. (2018). Scientific polarization. *European Journal for Philosophy of Science*, 8(3), 855–875.
- O'Reilly, C., Maher, P. J., & Quayle, M. (2024). The likes that bind: Even novel opinion sharing can induce opinion-based identification. *British Journal of Social Psychology*, 63(4), 2031–2051.
- Parsegov, S. E., Proskurnikov, A. V., Tempo, R., & Friedkin, N. E. (2017). Novel multidimensional models of opinion dynamics in social networks. *IEEE Transactions on Automatic Control*, 62(5), 2270–2285.
- Pérez-Martínez, H., Minguez, F. B., Soriano-Paños, D., Gómez-Gardeñes, J., & Floría, L. M. (2023). Polarized opinion states in static networks driven by limited information horizons. *Chaos Solitons & Fractals*, 175(1), Article 113917.
- Prasetya, H. A., & Murata, T. (2020). A model of opinion and propagation structure polarization in social media. *Computational Social Networks*, 7, 1–35.
- Proietti, C., & Chiarella, D. (2023). The role of argument strength and informational biases in polarization and bi-polarization effects. *The Journal of Artificial Societies and Social Simulation*, 26(2), 5.
- Proskurnikov, A. V., Matveev, A. S., & Cao, M. (2015). Opinion dynamics in social networks with hostile camps: Consensus vs. polarization. *IEEE Transactions on Automatic Control*, 61(6), 1524–1536.
- Quante, L., Stechemesser, A., Hödtke, D., & Levermann, A. (2024). Convolution of individual and group identity: Self-reliance increases polarisation in basic opinion model. *Humanities & Social Sciences Communications*, 11(1), 838.
- Quattrociocchi, W., Caldarelli, G., & Scala, A. (2014). Opinion dynamics on interacting networks: Media competition and social influence. *Scientific Reports*, 4, 4938.
- Racz, M. Z., & Rigobon, D. E. (2023). Towards consensus: Reducing polarization by perturbing social networks. *IEEE Transactions on Network Science and Engineering*, 10(6), 3450–3464.
- Rahaman, I., & Hosein, P. (2021). A model for optimizing article recommendation for reducing polarization. In *Proceedings of the IEEE/ACM international conference on advances in social networks analysis and mining* (pp. 107–110). Electr Network: Assoc Computing Machinery.
- Rao, V. R., & Steckel, J. H. (1991). A polarization model for describing group preferences. *Journal of Consumer Research*, 18(1), 108–118.
- Rathje, S., Van Bavel, J. J., & van der Linden, S. (2021). Out-group animosity drives engagement on social media. *Proceedings of the National Academy of Sciences of the United States of America*, 118(26), Article e2024292118.
- Rogowski, J. C., & Sutherland, J. L. (2016). How ideology fuels affective polarization. *Political Behavior*, 38(2), 485–508.
- Rychwalska, A., & Roszczyńska-Kurasinska, M. (2018). Polarization on social media: when group dynamics leads to societal divides. In *Proceedings of the 51st annual Hawaii international conference on system sciences* (pp. 2088–2097). Hi: Hicss.
- Santos, F. P., Lelkes, Y., & Levin, S. A. (2021). Link recommendation algorithms and dynamics of polarization in online social networks. *Proceedings of the National Academy of Sciences of the United States of America*, 118(50), Article e2102141118.
- Sarkar, A., & Hadfield, G. K. (2024). *Dynamics of polarization under normative institutions and opinion expression stewarding*. CoRR.
- Sasahara, K., Chen, W., Peng, H., Ciampaglia, G. L., Flammini, A., & Menczer, F. (2021). Social influence and unfollowing accelerate the emergence of echo chambers. *Journal of Computational Social Science*, 4(1), 381–402.
- Schweighofer, S., & Garcia, D. (2024). Raising the spectrum of polarization: Generating issue alignment with a weighted balance opinion dynamics model. *The Journal of Artificial Societies and Social Simulation*, 27(1), 5323.
- Selleck, S. (2022). Reducing polarization in opinion networks in the presence of stubborn leaders. *Chaos, Solitons & Fractals*, 16, 113–128.

- Shahbazi, M., & Bunker, D. (2024). Social media trust: Fighting misinformation in the time of crisis. *International Journal of Information Management*, 77, Article 102780.
- Shi, L., Li, W. H., Shi, M. J., Shi, K. B., & Cheng, Y. H. (2023). Opinion polarization over signed social networks with quasi-structural balance. *IEEE Transactions on Automatic Control*, 68(11), 6867–6874.
- Shi, F., Teplitskiy, M., Duede, E., & Evans, J. A. (2019). The wisdom of polarized crowds. *Nature Human Behaviour*, 3(4), 329–336.
- Sieber, J., & Ziegler, R. (2019). Group polarization revisited: A processing effort account. *Personality and Social Psychology Bulletin*, 45(10), 1482–1498.
- Sirbu, A., Pedreschi, D., Giannotti, F., & Kertész, J. (2019). Algorithmic bias amplifies opinion fragmentation and polarization: A bounded confidence model. *Plos One*, 14(3), Article e0213246.
- Sobkowicz, P. (2018). Opinion dynamics model based on cognitive biases of complex agents. *The Journal of Artificial Societies and Social Simulation*, 21(4), 8.
- Sobkowicz, P. (2023). Social depolarization and diversity of opinions-unified ABM framework. *Entropy*, 25(4), 568.
- Stewart, A. J., McCarty, N., & Bryson, J. J. (2020). Polarization under rising inequality and economic decline. *Science Advances*, 6(50), eabd4201.
- Stewart, A. J., Plotkin, J. B., & McCarty, N. (2021). Inequality, identity, and partisanship: How redistribution can stem the tide of mass polarization. *Proceedings of the National Academy of Sciences of the United States of America*, 118(50), Article e2102140118.
- Stoner, J. A. F. (1961). *A comparison of individual and group decisions involving risk*. Massachusetts Institute of Technology.
- Sun, R. A., Zhu, H., & Guo, F. (2023). Impact of content ideology on social media opinion polarization: The moderating role of functional affordances and symbolic expressions. *Decision Support Systems*, 164, Article 113845.
- Sunstein, C. R. (2001). Why they hate us: The role of social dynamics. *Harvard Journal of Law and Public Policy*, 25(2), 429–440.
- Sunstein, C. R. (2002). The law of group polarization. *Journal of Political Philosophy*, 10(2), 175–195.
- Tao, Y. Q., Hu, B., Zeng, Z. L., & Ma, X. M. (2025). Detecting and regulating sentiment reversal and polarization in online communities. *Information Processing and Management*, 62(1), Article 103965.
- Tokita, C. K., Guess, A. M., & Tarnita, C. E. (2021). Polarized information ecosystems can reorganize social networks via information cascades. *Proceedings of the National Academy of Sciences of the United States of America*, 118(50), Article e2102147118.
- Törnberg, P. (2022). How digital media drive affective polarization through partisan sorting. *Proceedings of the National Academy of Sciences of the United States of America*, 119(42), Article e2207159119.
- Turner, M. A., & Smaldino, P. E. (2018). Paths to polarization: How extreme views, miscommunication, and random chance drive opinion dynamics. *Complexity*, 2018(1), Article 2740959.
- Valensise, C. M., Cinelli, M., & Quattrociocchi, W. (2023). The drivers of online polarization: Fitting models to data. *Information Sciences*, 642, Article 119152.
- van der Linden, S., Leiserowitz, A., Rosenthal, S., & Maibach, E. (2017). Inoculating the public against misinformation about climate change. *Global Challenges*, 1(2), Article 1600008.
- Vasist, P. N., Chatterjee, D., & Krishnan, S. (2024). The polarizing impact of political disinformation and hate speech: A cross-country configurational narrative. *Information Systems Frontiers*, 26(2), 663–688.
- Waller, I., & Anderson, A. (2021). Quantifying social organization and political polarization in online platforms. *Nature*, 600(7888), 264–268.
- Wang, L., Wang, C., & Yao, X. Y. (2024). Befriended to polarise? The impact of friend identity on review polarisation-A quasi-experiment. *Information Systems Journal*, 34(3), 680–710.
- Webster, S. W., & Albertson, B. (2022). Emotion and politics: Noncognitive psychological biases in public opinion. *Annual Review of Political Science*, 25(1), 401–418.
- Webster, S. W., & Abramowitz, A. I. (2017). The ideological foundations of affective polarization in the US electorate. *American Politics Research*, 45(4), 621–647.
- Wen, T., Zheng, R., Wu, T., Liu, Z. Y., Zhou, M., Syed, T. A., Ghataoura, D., & Chen, Y. W. (2025). Formulating opinion dynamics from belief formation, diffusion and updating in social network group decision-making: Towards developing a holistic framework. *European Journal of Operational Research*, 325(3), 381–399.
- Weron, T., & Szabinski, J. (2022). Opinion evolution in divided community. *Entropy*, 24(2), 185.
- Williams, H. T. P., McMurray, J. R., Kurz, T., & Lambert, F. H. (2015). Network analysis reveals open forums and echo chambers in social media discussions of climate change. *Global Environmental Change-Human and Policy Dimensions*, 32, 126–138.
- Wu, Y., Li, L. J., Yu, Q. N., Gan, J. X., & Zhang, Y. (2023). Strategies for reducing polarization in social networks. *Chaos Solitons & Fractals*, 167, Article 113095.
- Wu, Z. B., Zhou, Q. Y., Dong, Y. C., Xu, J. P., Altalhi, A. H., & Herrera, F. (2022a). Mixed opinion dynamics based on DeGroot model and Hegselmann–Krause model in social networks. *IEEE Transactions on Systems, Man, Cybernetics: Systems*, 53(1), 296–308.
- Wu, J. S. T., Hauert, C., Kremen, C., & Zhao, J. Y. (2022b). A framework on polarization, cognitive inflexibility, and rigid cognitive specialization. *Frontiers in Psychology*, 13, Article 776891.
- Xia, H. X., Wang, H. L., & Xuan, Z. G. (2011). Opinion dynamics: A multidisciplinary review and perspective on future research. *International Journal of Knowledge Systems Science*, 2(4), 72–91.
- Xie, Z. H., Li, S. S., & Li, W. (2025). Do discussions in human-computer communities trigger group polarization? Insights from the media evocation paradigm. *Computers in Human Behavior*, 165, Article 108561.
- Xie, L., Wang, D. D., & Ma, F. C. (2023). Analysis of individual characteristics influencing user polarization in COVID-19 vaccine hesitancy. *Computers in Human Behavior*, 143, Article 107649.
- Xing, Y. F., Wang, X. W., Qiu, C. C., Li, Y. Q., & He, W. (2022). Research on opinion polarization by big data analytics capabilities in online social networks. *Technology in Society*, 68, Article 101902.
- Xing, Y. F., Zhang, J. Z., Teng, G. Q., & Zhou, X. T. (2024). Voices in the digital storm: Unraveling online polarization with ChatGPT. *Technology in Society*, 77, Article 102534.
- Yang, B. Z., Yu, Q., & Fan, Y. (2022). A hybrid opinion formation and polarization model. *Entropy*, 24(11), 1692.
- Ye, Y. J., Zhang, R. J., Zhao, Y. Q., Yu, Y. Y., Du, W. X., & Chen, T. G. (2022). A novel public opinion polarization model based on BA network. *Systems*, 10(2), 46.
- Yuan, J. J., Shi, J. W., Che, J. Y., Xu, C. H., & Wang, J. (2020). Modeling and simulation analysis of public opinion polarization in a dynamic network environment. *Concurrency and Computation-Practice & Experience*, 32(19), e5771.
- Yuan, J. J., Shi, J. W., Wang, J., & Liu, W. N. (2022). Modelling network public opinion polarization based on SIR model considering dynamic network structure. *Alexandria Engineering Journal*, 61(6), 4557–4571.
- Zafeiris, A. (2022). Opinion polarization in human communities can emerge as a natural consequence of beliefs being interrelated. *Entropy*, 24(9), 1320.
- Zhang, X. L., Akyol, E., & Ertem, Z. (2024). Polarization game over social networks. In *Proceedings of the ICC 2024-IEEE international conference on communications* (pp. 1–6).
- Zhang, B. W., Huang, W. J., Dong, Y. C., & Pedrycz, W. (2025). Utility-driven minimum cost consensus model with preference learning and fairness concerns. *IEEE Transactions on Systems, Man, Cybernetics: Systems*. <https://doi.org/10.1109/TSMC.2025.3552113>
- Zmigrod, L. (2020). The role of cognitive rigidity in political ideologies: Theory, evidence, and future directions. *Current Opinion in Behavioral Sciences*, 34, 34–39.